

VLST after ACS-PCI

Title, authors, abstract, keywords

Numerical source of truth: `paper/frozen_results.yaml`. Trace tags: `docs/paper_evidence_map.md` (W1–W5, B3, B11, §4.6, §5.8–5.9, §7.3). Live nested key: `nested_cv_v35_antileakage_on`. Do not quote excluded unlabeled nested PR-AUC 0.9771 / 0.9635, Version 4 0.8553 / 0.6742, or evidence-map §7.1 historical rows.

Title

VLST after ACS-PCI — association, nested-CV prediction, and TabPFN attribution on the Wang 2020 derivation cohort

Authors

Authors are **not listed** in `paper/frozen_results.yaml`. Complete investigator names, affiliations, and corresponding author here before journal submission. This placeholder is not a freeze scalar.

Structured abstract

Background

Very late stent thrombosis (VLST) is Academic Research Consortium 2007 definite stent thrombosis more than one year after percutaneous coronary intervention (PCI). It is rare (92 events among 5,185 patients in this file; prevalence 0.0177) and analysed here as a binary label on Wang 2020's derivation cohort. Wang published an 8-variable Cox integer score on these same rows (published derivation c-statistic 0.80 [0.75, 0.85]; Shantou external c-statistic 0.82 for **Wang's score**, file not in this repository). Classic tabular classifiers on this file are easy to inflate if follow-up time remains in the matrix. Tabular prior-fitted networks (TabPFN) offer in-context learning on mixed-type tables without a per-dataset hyperparameter grid; this paper ranks four TabPFN arms and five class-weighted **library-default** tree/linear models under a shared nested-CV protocol after anti-leakage drops. The nine-arm table is that ranking, not a tuned-boosting contest.

Methods

Retrospective derivation-cohort analysis of VLST.csv (The First Hospital of Jilin University; ACS, age ≥ 18 years, PCI 1 January 2014 – 1 June 2015; $n = 5185$). Outcome = binary Stent thrombosis. Nested evaluation implements ALL LEAKS OFF: drop identifiers, TSSI (mixed time-to-event vs completed follow-up), and WBC (recording-precision / batch marker); quantize named labs (Cre/CaI/Fibrinogen/Fast-Glu) **before** splitting; fit the stent-brand codebook on the **train fold only**; no SMOTE; clone scaler/OHE inside splits. Prediction uses stratified nested 5 outer $\times$ 4 inner CV (seed 42); the inner loop tunes the F1 threshold only. Five classic arms (logistic regression, random forest, XGBoost, LightGBM, CatBoost) are an **untuned reference panel** (library defaults plus class weighting); GridSearchCV winners from 70/30 leakage twins are **not** imported. Four TabPFN arms: local v3 / v3.5 (tabpfn==9.0.0) and hosted thinking-high v3 / v3.5 (tabpfn-client==0.6.0; effort high; metric average_precision). Primary ranking metric is PR-AUC at prevalence 0.0177. Uncertainty on pooled out-of-fold scores: stratified patient-level bootstrap ($n_{boot}=2000$, seed 42). Attribution uses two notebooks: 8-seed forward SFS, mutual information, and PDP on train; held-out SHAP / k-SII (same OFF cleaning).

Results

Among nine nested-CV arms (anti-leakage ON), TabPFN thinking v3.5 had PR-AUC 0.9212 [0.8785, 0.9613], ROC-AUC 0.9963, Brier 0.0047; nested recall (sensitivity) 0.8370, precision 0.8851, F1 0.8603 (5083/10/15/77). Local TabPFN v3.5 had PR-AUC 0.8957 [0.8446, 0.9426], Brier 0.0048; nested recall 0.7826, F1 0.8229. Highest library-default classic PR-AUC was XGBoost 0.6322 [0.5331, 0.7247], then LightGBM 0.6271 [0.5313, 0.7202] (nested F1 0.5818 and 0.5902). Paired bootstrap Δ PR-AUC thinking v3.5 – LightGBM was 0.2941 (0.2071–0.3805), $P(\Delta \leq 0) = 0/2000$. The frozen Wang integer score on the same 5,185 rows had ROC-AUC 0.8013 and PR-AUC 0.1032 (weights not re-fit). Table 4b adjusted ORs (Wald 95% CI): 1.1:1Post dilation 0.152 [0.081, 0.286]; Clopidogrel 0.480 [0.293, 0.787]; WBC 1.972 [1.667, 2.331] — associated with recorded VLST, not treatment effects.

Conclusions

On this derivation cohort, thinking v3.5 had the highest nested-CV PR-AUC among nine fitted arms (0.9212); local TabPFN v3.5 (0.8957) also exceeded library-default XGBoost (0.6322) and LightGBM (0.6271). That ranking is not a hyperparameter-matched contest against tuned boosting. TSSI, WBC, raw lab decimals, a full-cohort stent codebook, and train SMOTE must not enter nested-CV binary classifiers as if they were baseline covariates. Association findings are not treatment effects. Nested-CV discrimination on the derivation file is not external validation; the ML models were not externally or temporally tested.

Keywords

very late stent thrombosis; TabPFN; nested cross-validation; data leakage; class imbalance; acute coronary syndrome; percutaneous coronary intervention; SHAP

Keywords are topical, not freeze scalars.

Introduction

Very late stent thrombosis (VLST) is Academic Research Consortium 2007 *definite* stent thrombosis more than one year after implantation, angiographically confirmed. Probable and possible stent thrombosis are not counted. The clinical problem is late and rare: stent thrombosis accounts for a substantial share of new myocardial infarction after index PCI and, in Wang’s review, carries several-fold higher adjusted mortality than infarction unrelated to a previously stented site. The intended decision window is risk stratification **more than one year** after PCI, after the mandated dual-antiplatelet year.

The analysed file is Wang 2020’s derivation cohort: consecutive ACS patients aged ≥ 18 years undergoing PCI at The First Hospital of Jilin University, 1 January 2014 – 1 June 2015; 6,038 eligible $\rightarrow$ 5,185 analysed (236 in-hospital deaths, 413 refused follow-up, 204 lost — **cited from Wang**, not reconstructed here); **92** definite VLST events (prevalence **0.0177** / 1.77%). Median follow-up is 1,502 days; median PCI $\rightarrow$ VLST among events is 697 days. This is not an empty prediction field. The Dangas late stent thrombosis score had c-statistic 0.66 in **Wang’s** comparison (not recomputed here). Wang derived an 8-variable Cox VLST score on these same 5,185 rows (diabetes, previous PCI, AMI as admitting diagnosis, eGFR < 90 , 3-vessel disease, stents per lesion, SES, no post-dilation) with derivation c-statistic **0.80** [0.75, 0.85] and **Shantou** external c-statistic **0.82** ($n = 2,058$). That Shantou file is **not** in this repository (B11). Any claim that “no VLST score exists” is false.

Three limitations of that prior score, and of naive tabular ML on the same file, motivate the present analysis. First, Wang modelled **time-to-event** with follow-up as the time axis. Recoded as a binary classifier, Time since stent implantation (TSSI) is time-to-event for the 92 cases (minimum 380 days) and completed event-free follow-up for the 5,093 non-events (minimum 1,241 days). A rule “time $< 1,241 \rightarrow$ event” has zero false positives among controls. TSSI is **leakage**, not a baseline covariate (W1). Second, the raw file is **sorted by outcome** and cases were transcribed at a different numeric precision, so recording precision is a case/control batch marker (WBC; lab decimals). A signature-only probe (243 grid indicators, no magnitudes) reaches AP 0.4270. Wang excluded WBC from the Cox score because infection could not be ruled out; an FDR screen still ranks WBC (dual-label). Third, 92 events among 81 candidate columns is extreme class imbalance and low events-per-variable ($EPV \approx 1.14$ on the candidate list; ≈ 7.1 on the identified 13-covariate logit). ROC-AUC is dominated by true negatives; PR-AUC at prevalence 0.0177 is the informative ranking scale.

Gradient-boosted trees and penalised logistic regression are the usual tabular baselines. On this file they are **easy to overstate**: a 70/30 GridSearch that keeps TSSI, WBC, unquantized labs, a full-cohort stent encoder, and train SMOTE inflates hold-out PR-AUC by 0.30–0.61 relative to the inverted flags (Results 4.1). Nested CV therefore leaves the five classic arms at library defaults plus class weighting (inner loop = F1 **threshold** only; twin GridSearch winners **not** imported). They are an **untuned reference panel** next to four TabPFN arms that have no per-dataset grid. The nine-arm table ranks those fitted objects; it is not “best XGBoost vs TabPFN” (W3.8).

TabPFN (Prior-Fitted Networks) is a tabular foundation model: in-context learning on mixed-type tables, native categoricals, no per-dataset grid. VLST here is a small- n , mixed-type problem (92 events, 81 raw columns after IDs). Pins for this manuscript: `tabpfn==9.0.0` (local v3 and v3.5 checkpoints) and `tabpfn-client==0.6.0` (hosted

thinking-high v3 / v3.5). v3 and v3.5 are different weights; thinking-high and local inference are different objects and are never collapsed (W3.4–W3.5). This pack does **not** claim that TabPFN is ready for clinical use, that nested CV is external validation, or that attributions are treatment effects (W5).

What this manuscript adds on the same derivation cohort, beyond Wang’s integer score: (i) a five-flag anti-leakage protocol and a twin 70/30 demonstration that those flags **together** inflate classic GridSearch ranking; (ii) nested 5×4 stratified CV ranking four TabPFN arms (v3 vs v3.5; thinking vs local) against five **library-default** classic arms under ALL LEAKS OFF — not a tuned-boosting contest; (iii) a frozen scoring of Wang’s published integer points on these 5,185 rows (historical comparator, not a nested arm); (iv) association screens (Table 4b) and TabPFN attribution (8-seed SFS, MI, PDP, held-out SHAP / *k*-SII) that **do not** feed the predictor. **What it does not add:** external or temporal testing of the ML models; a Cox linear-predictor re-fit; decision-curve analysis (B11); nested inner-loop GridSearch for classics.

Methods

Language follows pack terminology (**W5**): **association** for full-cohort univariate and multivariable screens; **prediction** only for nested-CV out-of-fold ranking; **interpretation** / **attribution** for selector catalogues and TabPFN explanations. Nested-CV discrimination on the derivation file is not external validation. Adjusted odds ratios and attribution maps are not treatment effects. Wang 2020 is a historical comparator on the same rows.

3.1 Study cohort and data preprocessing

This is a retrospective derivation-cohort analysis of `data/raw/VLST.csv` (Wang 2020 derivation file): consecutive ACS patients aged ≥18 years undergoing PCI at The First Hospital of Jilin University, 1 January 2014 – 1 June 2015. Wang reported 6,038 eligible and 5,185 analysed (236 in-hospital deaths, 413 refused follow-up, 204 lost). Those flow counts are cited from Wang 2020. Analysed *n* = **5,185**; **92** definite VLST events; 5,093 non-events; prevalence **0.0177**. Median follow-up 1,502 days; median PCI → VLST 697 days. Ethics NO. 2013-256, written informed consent, and NCT03491891 are cited from Wang. This repository analysis was not separately pre-registered.

Outcome: binary **very late stent thrombosis** (column `Stent thrombosis`) — ARC 2007 definite stent thrombosis more than one year after implantation, angiographically confirmed. Probable and possible stent thrombosis are not counted. Wang analysed time-to-event; this pack uses the stored 0/1 label (limitation **W3.2**, not a result).

Exploratory data analysis (`code/analyzes/eda.ipynb`) found **no missing values**. Identifiers `NO.` and `Name` are dropped. All later parts use these 5,185 rows; **splits differ** (full-cohort association; inner 80/20 selectors; nested 5×4 prediction; 70/30 leakage twins and attribution).

Quantization is file-level and pre-split. Rounding is a cleaning rule on named labs, not a parameter fit on *y*, and is applied **before** any train/test or CV split. Incentive: recording precision in `VLST.csv` is a case/control batch marker (raw file sorted by outcome; cases

transcribed at a different numeric convention). A signature-only probe (243 grid indicators, no magnitudes) reaches AP **0.4270** / ROC-AUC **0.9522**. Grid:

CLINICAL_QUANTIZE_PLACES = {Cre: 0, CaI: 2, Fiberinogen: 1, Fast-Glu: 1}

Fiberinogen is the CSV spelling (Cre integer $\mu\text{mol/L}$; CaI 2 dp conventional TnI — do not round to 0; Fiberinogen / Fast-Glu 1 dp). This is not post-split leakage.

3.2 Strict anti-leakage protocol

Operational definition = **five flags plus companion controls**, not “drop TSSI and WBC.” Nested CV, Part 2 selectors, and Part 5 attribution use the **ALL LEAKS OFF** state. Canonical pack copy: `paper_results/anti_leakage_protocol.md`. Freeze flow: drop NO. / Name / TSSI / WBC; quantize the four labs; stent codebook train-fold only; no SMOTE.

Flag	OFF (prediction / attribution / selectors)	ON (leakage demonstration only)	Incenti
KEEP_TSSI	False — drop Time since stent implantation	True	Mixed tin definition (time-to-event vs complete follow-up Rule “tim 1,241 → event” ha zero cont false positives. Recordin precision batch marker; Wang excluded WBC fro the Cox score (infection FDR still ranks it (dual-lab
DROP_WBC	True — drop WBC	False	Spurious decimals fingerprir batch/sou After OF cleaning,
QUANTIZE_CLINICAL	True (pre-split, §3.1)	False	

Flag	OFF (prediction / attribution / selectors)	ON (leakage demonstration only)	Incentive
			equalising leftover precision still drops TabPFN v3.5 nests AP by 0.0518 .
STENT_ENCODER_TRAIN_ONLY	True — encode_stent_on_fold	False (full-cohort codebook)	Rare brand must not define level from held-out rows. Unseen — Other.
USE_SMOTE	False	True in the ON twin only	Unmatched train synthesis; nested CV never use SMOTE.

TSSI is time-at-risk / completed follow-up, not a baseline covariate (VLST=1: time to thrombosis, min 380 days; VLST=0: event-free follow-up, min 1,241, max 1,605 days).

WBC is dropped from the ML matrix as a leaky post-event / recording-precision variable. Identifiers NO. and Name are dropped. Follow-up drugs (Aspirin, Clopidogrel, Ticagrelor, DAPT) **stay in** with a post-baseline caveat — they are not leak-flag drops.

Stent-brand encoding. STENT_BRAND_COL = "Stent type-SES"; brands with count < min_count=30 collapse to "Other" (106 raw strings → 9 levels). Nested CV uses encode_stent_on_fold on each outer-training fold (**not** a full-frame codebook before the split). Classic models then clone imputer / scaler / one-hot **inside each CV split**; TabPFN arms see the 9-level frame natively. EDA found no missing values, so imputers are inert.

Selectors never touch a parked 70/30 test. baseline_feature_selections.ipynb splits once: fit 1 - INNER_VAL_SIZE, score INNER_VAL_SIZE=0.2 (RANDOM_STATE=42) → 4,148 / 74 events vs 1,037 / 18 events. USE_CACHE=False. Selector catalogues do not feed nested-CV prediction. Twin GridSearch best_params_ are **not** imported into nested CV.

The complementary **ALL LEAKS ON** twin exists only to demonstrate leakage on the same 70/30 GridSearch family. It is not a nested-CV arm, not TabPFN, and not “identical except TSSI.” Nested TabPFN ON vs OFF is **[RE-SOURCE]**.

Where a hold-out is required (leakage twins; Part 5), one stratified train_test_split is used: TEST_SIZE=0.30, RANDOM_STATE=42 → train **3,629 / 64 events**, held-out **1,556 / 28 events**. That split is **not** the prediction evaluation.

3.3 Model zoo

Classic models — two non-interchangeable regimes

(a) GridSearchCV (leakage twins only). Seven families: logistic regression, decision tree, random forest, Gaussian naïve Bayes, CatBoost, XGBoost, LightGBM. Shared 3,629/1,556 split; GRID_SCORING="average_precision"; StratifiedKFold(n_splits=5, shuffle=True, random_state=42). SMOTE on the ON-twin training set only. Executed best_params_ are a leakage-axis supplement, **not** nested-CV hyperparameters.

(b) Main benchmark (nested CV). Five classics — LR, RF, XGB, LGBM, CatBoost — plus four TabPFN arms (nine RUN_MODELS). Constructors: library defaults plus class weighting (class_weight="balanced"; CatBoost auto_class_weights="Balanced"; XGBoost scale_pos_weight from the outer-train fold). DT and GNB are not nested-CV arms. **GridSearch winners from (a) are not imported.** No SMOTE. The five classics are an **untuned reference panel**, not a nested inner GridSearch.

TabPFN v3.5 (local) and API thinking mode

Pins (run_manifest.json, Tesla T4): **tabpfn==9.0.0, tabpfn-client==0.6.0.** model_path="auto" still resolves to v3 in tabpfn 9.0.0, so checkpoints are named explicitly.

Arm	Runtime	Checkpoint	Thinking
TabPFN v3	local tabpfn	tabpfn-v3-classifier-v3_default.ckpt	off (n_estimators="auto")
TabPFN v3.5	local tabpfn	tabpfn-v3.5-20260909.safetensors	off
TabPFN thinking v3	hosted tabpfn_client	v3_default	thinking_mode=True, thinking_effort="high", thinking_metric="average"
TabPFN thinking v3.5	hosted tabpfn_client	v3.5_default	same THINKING_KWARGS

v3 versus v3.5 are different weights; thinking versus local are different objects. They are never collapsed. Thinking constructors are unchanged from the freeze rule. Client quota comments (as of 2026, free tier): **20 thinking fits, 50 million prediction cells per day.** Interpretability keeps MI, SFS, and PDP on local v3.5 (FS_THINKING_MODE=False; 0 client fits) so quota is reserved for held-out SHAP.

The published Wang 8-variable **integer score** is scored on all 5,185 rows with published Table 2 points **frozen** (weights not re-fit). It is not a nested-CV arm and not a Cox re-fit.

3.4 Evaluation protocol (honest tuning budget)

Primary ranking metric. Average precision (**PR-AUC**) at prevalence 0.0177. ROC-AUC and Brier are reported alongside and are not collapsed across TabPFN arms.

Nested cross-validation — the only prediction evaluation. Stratified nested CV, 5 **outer** / 4 **inner** folds, outer shuffle=True, random_state=42. Events per outer fold: 18, 18, 18, 19, 19 (**W4**). Inner CV tunes the **F1 decision threshold only** — not hyperparameters and not feature sets. **Single nested CV, not repeated nested CV.** “Repeated stratified splits” in this pack means: (i) four inner stratified folds per outer fold; (ii) eight shuffled seeds for stability SFS; (iii) 2,000 stratified bootstrap resamples of **stored** OOF scores (classifiers not re-fit). It does **not** mean repeated nested CV of the whole 5×4 scheme.

Honest nested operating point (quote Table 2). Per-fold inner-CV F1 threshold applied once to the unseen outer fold. Precision, recall (sensitivity), F1, F2 ($\beta = 2.0$), and 2×2 counts in Results 4.2 use this cut. A single pooled F1 cut on concatenated OOF labels is **optimistically biased** and is not the headline.

Bootstrap CIs. Patient-level stratified bootstrap of pooled nested OOF (n_boot=2000, seed 42), keeping 92 events and 5,093 non-events. Paired Δ PR-AUC uses the same resamples.

Events per variable (W4). 92/81 $\approx$ **1.14**; unidentified Table 4 92/17 $\approx$ **5.4** (not quoted as the association model); identified Table 4b 92/13 $\approx$ **7.1** (still below EPV ≥ 10). Association logits do not use class_weight. Firth on the same 13 covariates is **EDA-only** — not a nested-CV arm, never fused into GridSearch.

Unequal tuning (stated in the ranking claim). TabPFN has no per-dataset grid. Classics in nested CV are defaults plus class weights. The nine-arm table ranks those objects; it is not a hyperparameter-matched contest (**W3.8**).

3.5 Interpretability: 8-seed SFS, MI, PDP, held-out SHAP / k-SII

A full interpretability run exceeds the Kaggle session limit. Parent tabpfn_interpretability.ipynb is archived; live split:

1. **tabpfn_interpretability_fs_pdp.ipynb** — mutual_info_classif on **train**; stability forward SFS (STABILITY_N_SEEDS=8; a 10-seed plan was abandoned under the session limit); PDP; consensus inputs. Local TabPFN v3.5; **0 client fits**. Rankings use the 3,629-row train. Not a nested-CV feature mask. Live dump: 8/8 seeds completed; 3/10 provisional tables are **archived**.
2. **tabpfn_interpretability_shap.ipynb** — held-out SHAP (shapiq SV) and native SHAP-IQ (k-SII / waterfall). Intended constructor: client thinking (INTERP_THINKING_MODE=True, effort high, metric average_precision, v3.5). SHAP on **all 1,556 held-out rows**. k-SII / waterfall: first held-out VLST=1 row (cohort index **5176**; budget 256) — one-row interaction view, not a cohort screen. Stored dump: HTTP 429 then local finish of remaining SHAP rows (execution note, not a change of intended constructor).

Both notebooks apply the same ALL LEAKS OFF cleaning as nested CV (IDs + TSSI + WBC dropped; CLINICAL_QUANTIZE_PLACES; stent codebook train-only; 80 columns; no SMOTE). PDP is a train empirical curve near prevalence (PDP_USE_CLIENT=False), not nested-CV risk. Consensus ranking is a Borda-style mean of normalized ranks across train MI, train SFS frequency, and held-out mean([SHAP]).

Numerical SoT remains paper/frozen_results.yaml. Notebooks and Kaggle dumps are provenance.

Results

All prediction numbers below carry checkpoint / thinking / leak-status tags. Nested ranking is anti-leakage **ON** (ALL LEAKS OFF: IDs+TSSI+WBC dropped; labs quantized pre-split; stent codebook train-fold only; no SMOTE). Nested with vs without anti-leakage for TabPFN is **[RE-SOURCE]** — no matched 9-arm nested OFF dump. Do not quote excluded unlabeled nested PR-AUC 0.9771 / 0.9635. Language: association $\neq$ prediction $\neq$ attribution (**W5**).

The analysed file has **5,185** patients, **92** VLST events, **5,093** non-events, prevalence **0.0177**. Wang's flow (cited): 6,038 eligible $\rightarrow$ 5,185 analysed. No missing values. Cohort contrasts (association, not prediction) are **Table 1**. Example: previous PCI 1.85% vs 10.87% (Fisher $p = 1.25e-05$); eGFR 120.03 (34.10) vs 95.88 (19.63) (Welch $p = 4.64e-20$); 1.1:1Post dilation 49.01% vs 15.22% (χ^2 $p = 1.30e-10$). DAPT / Clopidogrel columns are follow-up persistence, not index-PCI prescriptions. LV and CaI remain unnamed. Identified multivariable screen (Table 4b, 13 covariates, EPV $\approx$ 7.1): 1.1:1Post dilation adj. OR **0.152** [0.081, 0.286]; Clopidogrel **0.480** [0.293, 0.787]; WBC **1.972** [1.667, 2.331]; Previous PCI **6.710** [2.884, 15.610]; eGFR **0.568** [0.449, 0.717]; LV **1.832** [1.539, 2.181]. OR < 1 is lower modelled odds of recorded VLST, not a treatment benefit. Quote Table 4b, not unidentified Table 4 (EPV $\approx$ 5.4).

Variable	No VLST (n=6065)	VLST (n=92)	Test	p
Age, years	59.83 (9.93)	60.71 (11.33)	Welch t	0.463
LVEF, %	55.15 (4.52)	54.55 (3.68)	Mann-Whitney U	0.033
LV (unnamed; not in Wang Table 1)	44.55 (4.04)	49.11 (4.23)	Welch t	5.44e-17
WBC, 10 ⁹ /L	8.75 (3.24)	12.49 (3.92)	Mann-Whitney U	7.90e-21
Creatinine	72.53 (24.81)	72.44 (19.05)	Mann-Whitney U	0.879
eGFR	120.03 (34.10)	95.88 (19.63)	Welch t	4.64e-20
Cal (unnamed; not in Wang Table 1)	37.37 (61.64)	40.55 (72.25)	Mann-Whitney U	0.051
Fibrinogen, g/L	3.17 (0.88)	3.37 (1.01)	Mann-Whitney U	0.012
Stents per lesion	1.21 (0.46)	1.42 (0.65)	Mann-Whitney U	0.000
Total stent length, mm	31.70 (15.62)	36.46 (20.71)	Mann-Whitney U	0.001
Men	3489 (68.51%)	68 (73.91%)	Chi-square	0.268
Diabetes	1293 (25.39%)	36 (39.13%)	Chi-square	0.003
Hypertension	2670 (52.42%)	51 (55.43%)	Chi-square	0.567
Dyslipidaemia (HL)	1612 (31.65%)	28 (30.43%)	Chi-square	0.804
Current smoker	2860 (56.16%)	54 (58.70%)	Chi-square	0.626
Previous PCI	94 (1.85%)	10 (10.87%)	Fisher	1.25e-05
Previous MI	347 (6.81%)	10 (10.87%)	Chi-square	0.128
Admitting diagnosis AMI	3095 (60.77%)	65 (70.65%)	Chi-square	0.054
3-vessel disease	1422 (27.92%)	42 (45.65%)	Chi-square	0.000
SES (SES column; matches Wang Table 1 SES)	3502 (68.76%)	76 (82.61%)	Chi-square	0.004
1:1:1 post-dilation (CSV as stored)	2496 (49.01%)	14 (15.22%)	Chi-square	1.30e-10
No postdilation (CSV as stored)	2597 (50.99%)	78 (84.78%)	Chi-square	1.30e-10
eGFR < 90 (CKD90)	860 (16.89%)	32 (34.78%)	Chi-square	6.55e-06
DAPT during follow-up (not index PCI)	2260 (44.37%)	35 (38.04%)	Chi-square	0.226

Table 1. Cohort characteristics

4.1 Baseline and anti-leakage contrast

Same seven classic models, same stratified 70/30 (train 3,629 / test 1,556, seed 42), GridSearchCV (GRID_SCORING=average_precision). **Not nested CV. Not TabPFN.** Five flags flip together; SMOTE is ON only in the leaks-on arm. **Incentives:** TSSI = mixed time-to-event vs completed follow-up; WBC = recording-precision / batch marker; unquantized labs = decimal-grid fingerprint (signature probe AP 0.4270); full-cohort stent codebook = test-brand leak; train SMOTE = unmatched inflation. Full protocol: `paper_results/anti_leakage_protocol.md`. Full tables: **Table 2** and `leakage_contrast_paper_figures_and_tables.md`.

Tag	Notebook	Flags
ALL LEAKS ON	<code>baseline_tssi_leakage.ipynb</code>	KEEP_TSSI=True, DROP_WBC=False, QUANTIZE_CLINICAL=False, STENT_ENCODER_TRAIN_ONLY=False, USE_SMOTE=True
ALL LEAKS OFF	<code>baseline_without_tssi.ipynb</code>	inverse flags, USE_SMOTE=False

Hold-out PR-AUC (primary ranking metric at prevalence 0.0177); Δ = ON – OFF:

Model	ON	OFF	Δ PR-AUC
Logistic regression	0.9134	0.3431	+0.5703
Decision tree	0.7524	0.1378	+0.6146
Random forest	0.9400	0.4874	+0.4526

Model	ON	OFF	Δ PR-AUC
Gaussian NB	0.2728	0.0564	+0.2163
CatBoost	0.9599	0.4942	+0.4656
XGBoost	0.9547	0.5685	+0.3862
LightGBM	0.9687	0.6675	+0.3011

Every model's hold-out PR-AUC is higher on ALL LEAKS ON. Inflation ranges **+0.30 to +0.61**. LightGBM is the least inflated booster and still drops by 0.30.

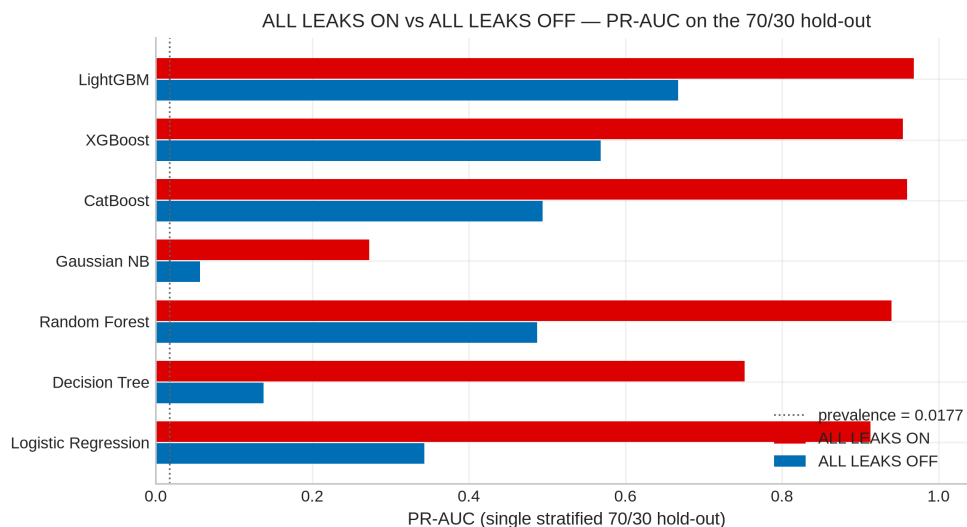


Figure S-TSSI. PR-AUC ALL LEAKS ON vs OFF

ROC-AUC compression is smaller. **Dual-label:** Gaussian NB ROC-AUC is slightly *higher* OFF (0.7493 vs 0.7425) while PR-AUC still falls (0.2728 $\rightarrow$ 0.0564). Do not write “every metric falls.”

At the notebook operating point, random forest F1 on ALL LEAKS OFF is **0** (no predicted events) while ROC-AUC remains 0.9287. Boosters that look near-perfect on ON (F1 0.9231, precision 1.0) drop to F1 0.41–0.55 once leak flags are off. Accuracy stays high except GNB because 1,528/1,556 hold-out rows are non-events; accuracy is not the ranking metric.

Reading. A 70/30 GridSearch pipeline that retains TSSI (and WBC, unquantized labs, a full-cohort stent encoder, and train SMOTE) **overstates hold-out ranking on this derivation file**. Nested CV, Part 2, and Part 5 therefore implement the OFF flags. This subsection does not transfer to TabPFN. **ARCHIVED:** LR PR-AUC 0.9575 $\rightarrow$ 0.5077.

4.2 Nested ranking: TabPFN checkpoints vs library-default reference panel

Dump:

Kaggle_baseline_plus_tabpfn_results/baseline_plus_tabpfn_results/
(tabpfn==9.0.0, tabpfn-client==0.6.0, Tesla T4, 9 arms). Nested 5 $\times$ 4, seed 42, inner loop = F1 threshold only. Five classic arms = **untuned reference panel** (defaults + class weighting; GridSearch winners **not** imported). No SMOTE. Leak status for the

entire table: anti-leakage **ON** (five-flag OFF: IDs+TSSI+WBC dropped; labs quantized pre-split; encode_stent_on_fold; no SMOTE). Full numbers: **Table 3**. Bootstrap 95% CIs: stratified resample of stored OOF, n_boot=2000, seed 42; models not re-fit.

Figure 1. Nested-CV out-of-fold PR and ROC curves.

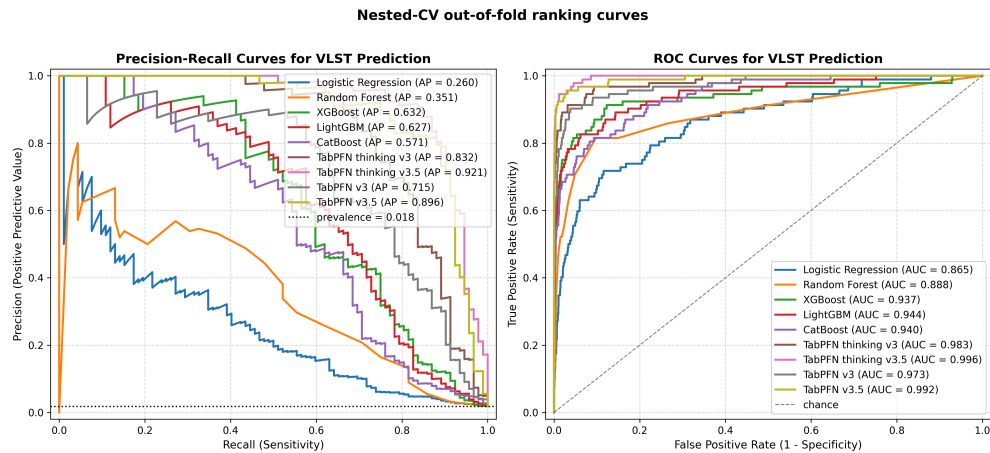


Figure 1. Nested OOF PR and ROC

Pooled nested OOF ranking (threshold-independent):

Rank	Model	Checkpoint	Thinking	PR-AUC (95% CI)	ROC-AUC	Brier
1	TabPFN thinking v3.5	hosted v3.5_default	thinking-high	0.9212 [0.8785, 0.9613]	0.9963	0.0047
2	TabPFN v3.5	local tabpfn-v3.5-20260909.safetensors	local	0.8957 [0.8446, 0.9426]	0.9916	0.0048
3	TabPFN thinking v3	hosted v3_default	thinking-high	0.8319 [0.7626, 0.8945]	0.9834	0.0060
4	TabPFN v3	local v3 ckpt	local	0.7150 [0.6260, 0.8074]	0.9731	0.0095
5	XGBoost	n/a	n/a	0.6322 [0.5331, 0.7247]	0.9374	0.0100
6	LightGBM	n/a	n/a	0.6271 [0.5313, 0.7202]	0.9444	0.0100
7	CatBoost	n/a	n/a	0.5707 [0.4750, 0.6729]	0.9404	0.0108
8	Random forest	n/a	n/a	0.3506 [0.2660, 0.4633]	0.8883	0.0150

Rank	Model	Checkpoint	Thinking	PR-AUC (95% CI)	ROC-AUC	Brier
9	Logistic regression	n/a	n/a	0.2596 [0.1834, 0.3588]	0.8651	0.0788

TabPFN vs library-default reference panel. Highest default-classic PR-AUC is XGBoost 0.6322, then LightGBM 0.6271. These are not GridSearch winners. Paired bootstrap Δ PR-AUC: thinking v3.5 – LightGBM **0.2941** (0.2071–0.3805), $P(\Delta \leq 0) = 0/2000$; TabPFN v3.5 – LightGBM **0.2686** (0.1822–0.3559), $P = 0/2000$; thinking v3.5 – XGBoost **0.2891** (0.2038–0.3791), $P = 0/2000$. The Δ is a contrast of stored OOF vectors, not evidence that tuned boosting would lose by the same margin.

v3 vs v3.5 (same thinking status). Local: 0.8957 vs 0.7150. Thinking: 0.9212 vs 0.8319.

Thinking vs local (same checkpoint family). v3.5: 0.9212 vs 0.8957. v3: 0.8319 vs 0.7150. Thinking v3.5 is **not** higher than local v3.5 in every fold (fold 1: local 0.9365 vs thinking 0.8802). Fold PR-AUC thinking v3.5: 0.8802, 0.9078, 0.9434, 0.9858, 0.8961.

Honest nested operating points (inner-fold F1 thresholds — quote these, not pooled cuts). Thinking v3.5: t 0.318 $\pm$ 0.059; 5083/10/15/77; sensitivity **0.8370**; precision **0.8851**; F1 **0.8603**. TabPFN v3.5: 5082/11/20/72; sensitivity **0.7826**; precision **0.8675**; F1 **0.8229**. XGBoost: sensitivity 0.5217; F1 0.5818. LightGBM: sensitivity **0.5870**; F1 **0.5902**. ROC-AUC CIs (thinking v3.5 **0.9963** [0.9932, 0.9986]; TabPFN v3.5 **0.9916** [0.9828, 0.9978]). Nested-CV discrimination on the derivation file is not external validation.



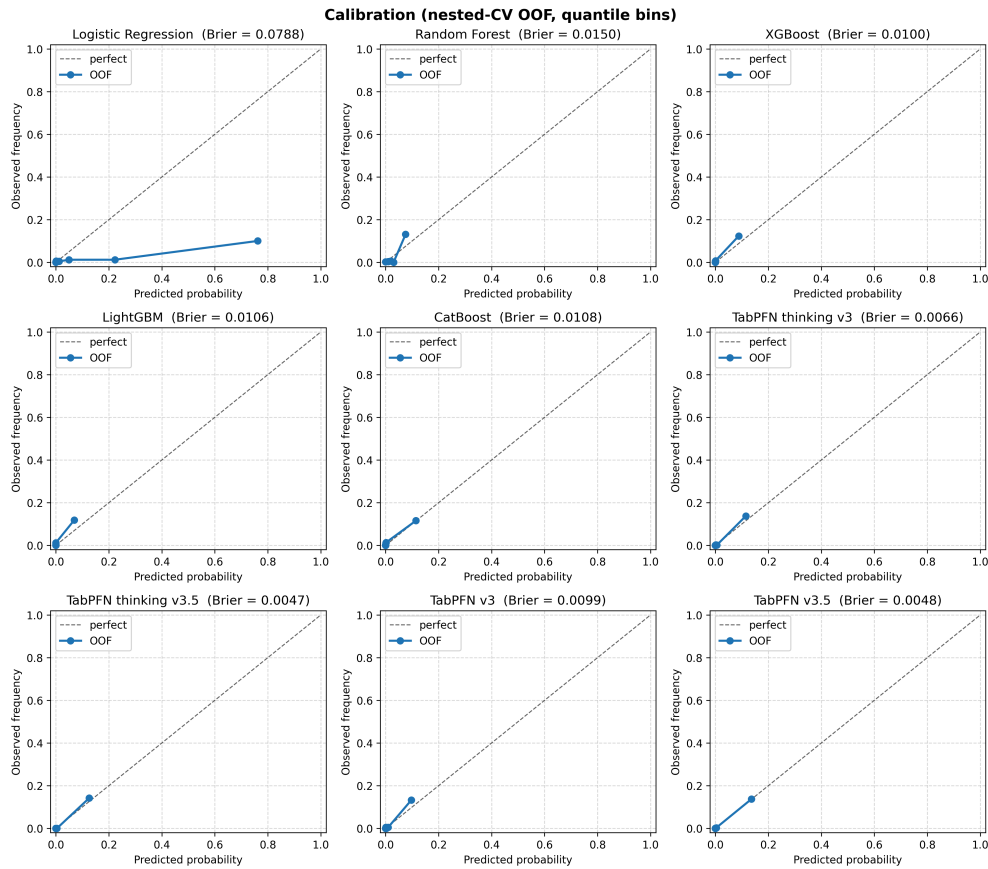


Figure 2. Nested calibration

Wang integer score (frozen comparator, not a nested arm). Full-cohort ROC-AUC **0.8013**, PR-AUC **0.1032**. SES points use PES; four post-dilation points use No postdilation = 1. Flipped encoding ROC-AUC **0.5084** is rejected.

4.3 Feature-selection stability across 8 seeds

Forward sequential feature selection (keep 10 of 80, 5-fold CV, average precision) repeated over **8/8 shuffled seeds** on the **train** split ($n = 3,629$), local TabPFN v3.5, FS_THINKING_MODE=False, ALL LEAKS OFF (same quantization + train-only stent codebook as nested CV). **Table 4.** Selected in **8/8**: CaI, LV, eGFR. Selected in 7/8: Cre, HbA1c. Selected in 6/8: Age. Selected in 5/8: LVEF. Selected in 4/8 (0.5 cutoff): 1.1:1Post dilation. WBC is not a column. A 10-seed plan was not finished; **3/10 provisional** tables are archived and are not this result.

Feature	Selected	Frequency
CaI	8/8	1.000
LV	8/8	1.000
eGFR	8/8	1.000
Cre	7/8	0.875
HbA1c	7/8	0.875
Age	6/8	0.750
LVEF	5/8	0.625
1.1:1Post dilation	4/8	0.500
LDL	2/8	0.250
No postdilation	2/8	0.250
TCL	2/8	0.250
Fibrinogen	1/8	0.125
Initial diagnosis-AMI	1/8	0.125
Previous PCI	1/8	0.125
Slow flow	1/8	0.125
Stent type-SES	1/8	0.125

Table 4. SFS stability (8/8 seeds)

Train mutual information (top 5 of 80): CaI 0.020536, LV 0.012818, eGFR 0.009424, LDL 0.009367, HbA1c 0.007328. Cre train MI is 0.000338 (rank 50 of 80). These catalogues are **attribution**, not a nested-CV feature mask.

Classic-selector overlap (Part 2/3, anti-leak 19-Sep dump, ML consensus n = 10 vs FDR-20): intersection {Clopidogrel, HbA1c, LV, No postdilation, eGFR}; Jaccard $5/25 = 0.20$. Dual-label: WBC is FDR-only.

4.4 Feature attribution and interactions (SHAP, *k*-SII, PDP)

Fit on train; SHAP explains **all 1,556 held-out rows** (client thinking-high intended constructor). Mean(|SHAP|) top 4 of 80: **eGFR 1.2288, CaI 1.0867, Cre 0.8093, LV 0.4828**. WBC is absent. Do not call this 15+15 or global SHAP on 5,185.

SHAP summary — feature impact on P[Stent thrombosis] (held-out, n=1556)

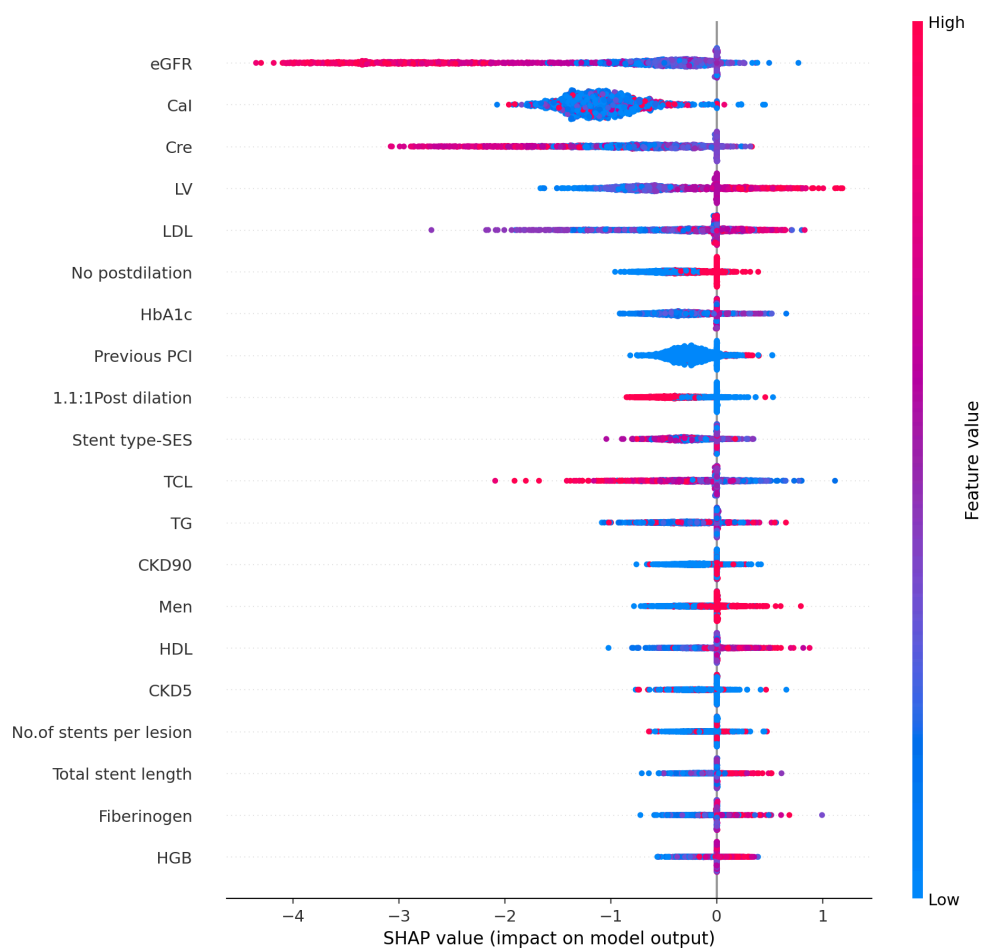


Figure. SHAP summary (held-out)

SHAP bar — mean(|SHAP|) on the held-out split (n=1556)

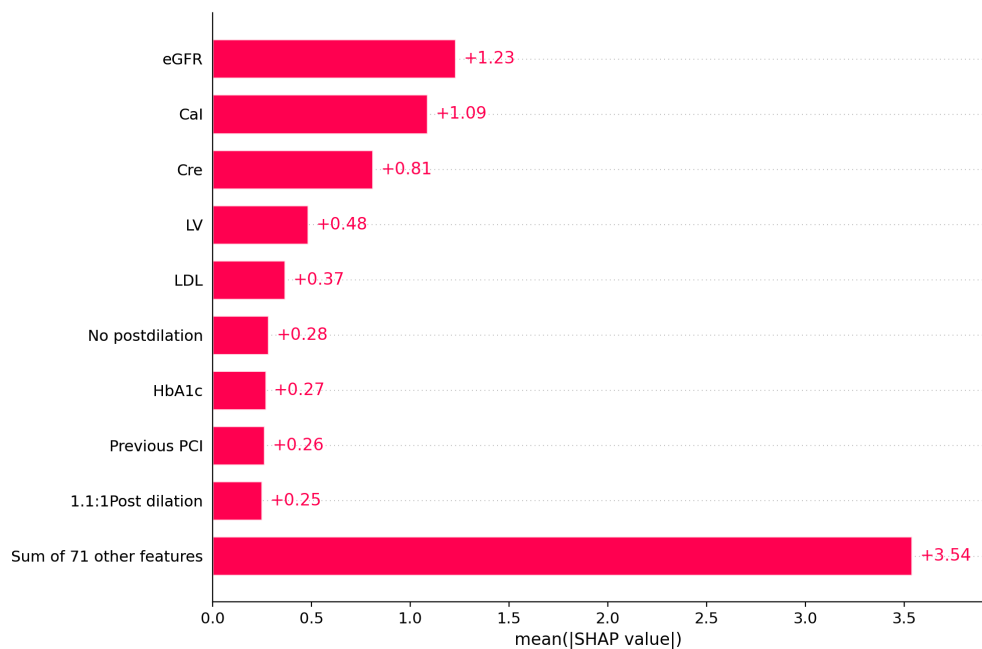


Figure. Mean |SHAP| bar

Borda consensus after merging held-out SHAP into the fs dump: **Cal, eGFR, LV** are 3/3 (also HbA1c and 1.1:1Post dilation at the 0.5 SFS cutoff).

PDP (local TabPFN, empirical prior, **train** n = 3,629). Y-axis near prevalence (~ 0.018). **Not** nested-CV risk. Largest binary $|\Delta|$: Previous PCI 0.017711 $\rightarrow$ 0.019006 (Δ **+0.001294**). 1.1:1Post dilation Δ **-0.000094**. A negative Δ is a lower modelled probability of recorded VLST, not a treatment benefit.

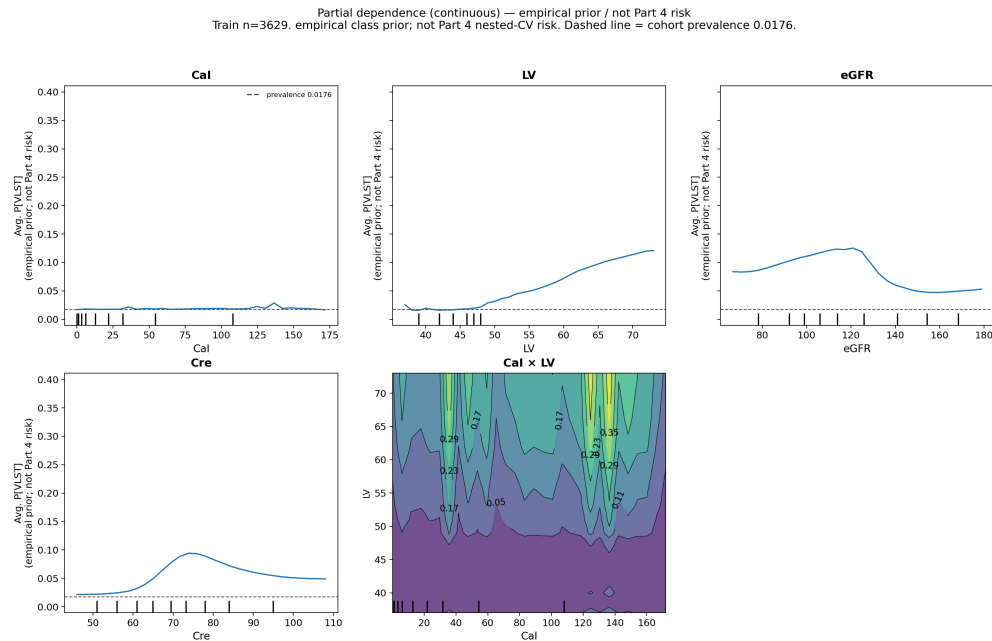


Figure. Continuous PDP (train, empirical prior)

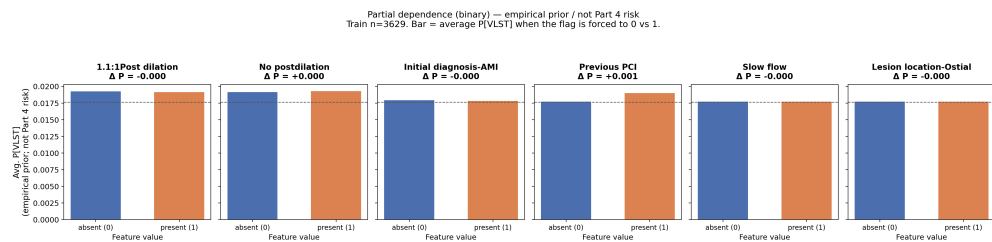


Figure. Binary PDP

k-SII / waterfall use one held-out VLST=1 row (cohort **5176**, budget 256). They illustrate how TabPFN combines features **for that patient**; they are not a cohort interaction screen.

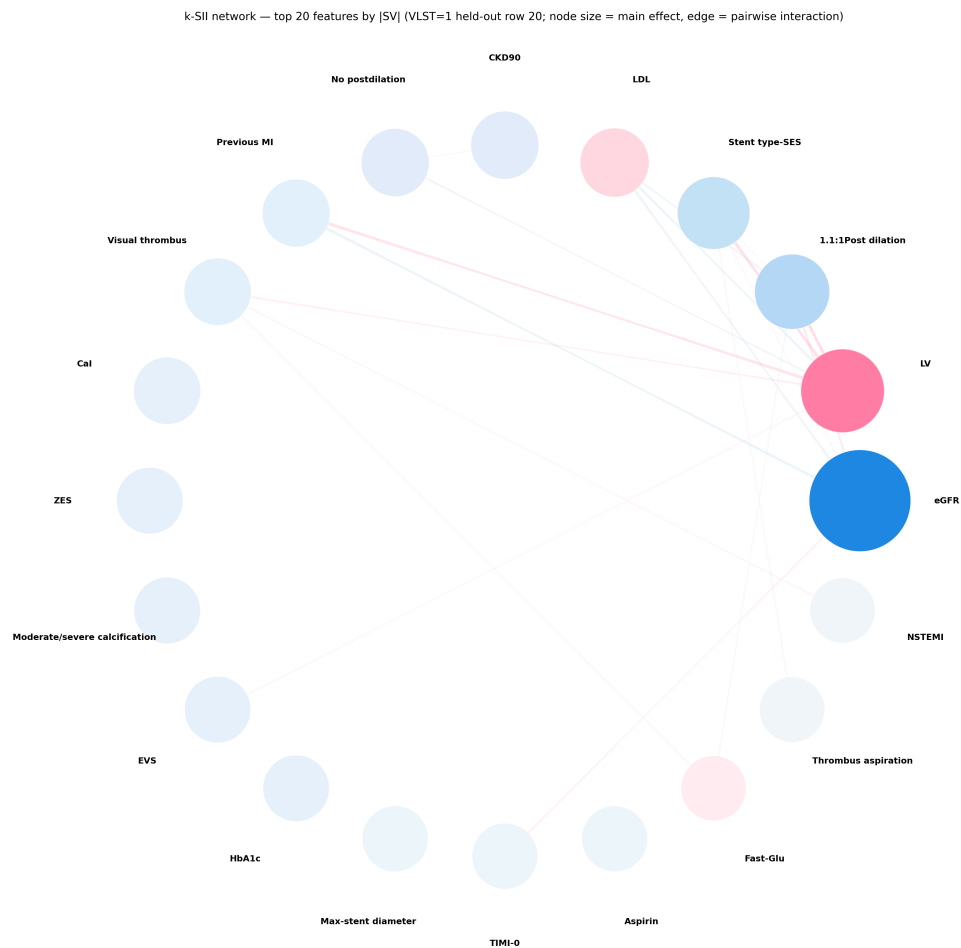


Figure. k-SII network (row 5176)

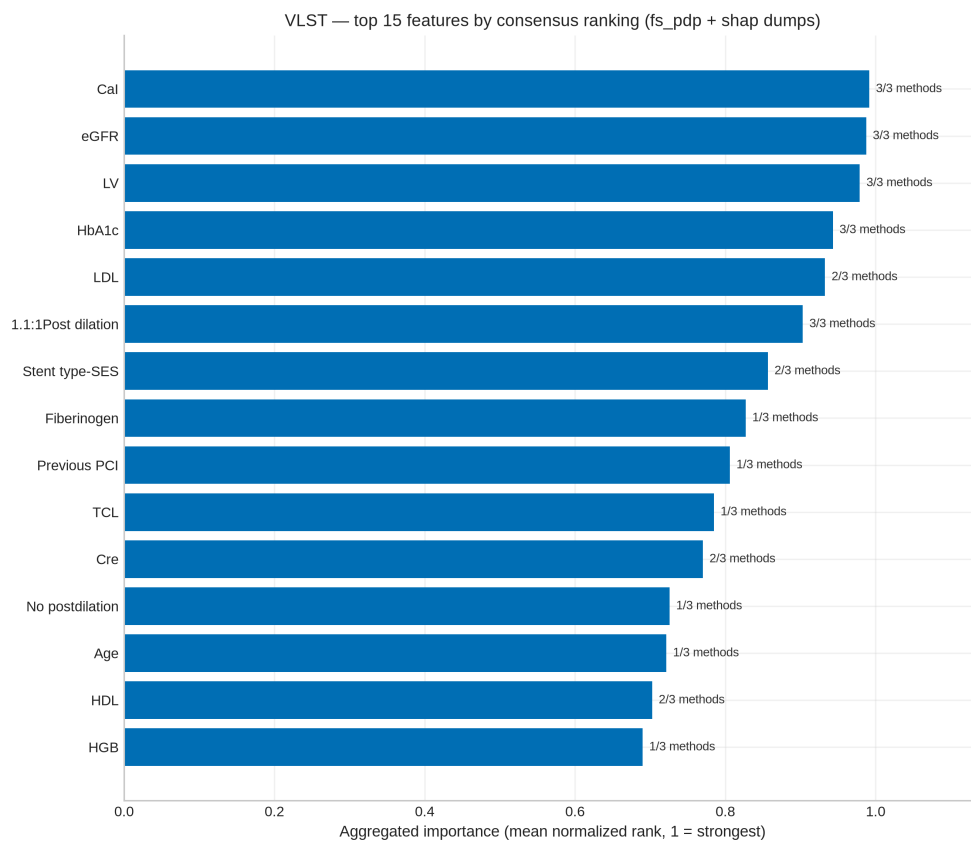


Figure. Consensus ranking

Discussion

Numbers are only those frozen in `paper/frozen_results.yaml`. Association $\neq$ prediction $\neq$ attribution (**W5**). No causal, “risk factor,” or clinical-utility claims. Decision-curve analysis is absent (**B11**).

5.1 Principal findings — TabPFN under extreme imbalance without a per-dataset grid

The ranking result that survives anti-leakage handling is derivation-cohort nested CV, not a bedside rule. After the full OFF protocol, TabPFN thinking v3.5 has the highest nested PR-AUC among nine fitted arms (**0.9212** [0.8785, 0.9613]); local TabPFN v3.5 is second (**0.8957** [0.8446, 0.9426]). Both exceed **library-default** XGBoost **0.6322** and LightGBM **0.6271** (paired Δ thinking v3.5 – LightGBM **0.2941** (0.2071–0.3805), $P(\Delta \leq 0) = 0/2000$). That is a ranking of these objects, not “TabPFN versus tuned boosting.” Nested F1 at inner-fold thresholds: thinking v3.5 **0.8603** (sensitivity 0.8370); local v3.5 **0.8229**; LightGBM **0.5902**.

Why TabPFN can rank above this **untuned reference panel** on this file is a design fact, not a mechanistic proof: it is a tabular foundation model (in-context learning, native categoricals, no per-dataset grid) on a small- n mixed-type table (92 events). Classics in the nested zoo are library defaults plus class weighting; the inner loop tunes only the F1 threshold. Unmatched search effort is part of the **claim** (W3.8), not only a limitation. v3 and v3.5 are different checkpoints; thinking-high (`tabpfn-client`, effort high) and local `tabpfn` are different objects. Thinking outranks local within each family on pooled PR-AUC, **but not in every fold** (fold 1: local v3.5 0.9365 vs thinking 0.8802). With 18–19 events per outer fold, fold-to-fold movement is expected. Ranking uses PR-AUC because prevalence is **0.0177**; ROC-AUC is high for every arm (0.865–0.996) and is the less informative scale.

What does **not** survive is treating a leaks-on 70/30 scoreboard, or the excluded unlabeled nested dump (thinking **0.9771** / local **0.9635**), as the TabPFN result. **[RE-SOURCE]**: there is no matched 9-arm nested OFF dump, so a numeric “nested v3.5 ON minus nested v3.5 OFF” cannot be written.

5.2 Clinical context — stent characteristics and biomarkers (association / attribution)

Names that recur after anti-leakage, as **associations or attributions**, not treatment effects:

- Table 4b (identified logit): post-dilation and Clopidogrel have adjusted OR < 1 ; WBC, previous PCI, and LV have OR > 1 . DAPT flags are follow-up persistence after the mandated year, not index-PCI prescriptions (W3.6).

- Part 5 stability SFS **8/8**: {CaI, LV, eGFR}. Held-out mean |SHAP|: eGFR 1.2288, CaI 1.0867, Cre 0.8093, LV 0.4828.
- Part 3 FDR $\cap$ ML three-way ($n = 5$): Clopidogrel, HbA1c, LV, No postdilatation, eGFR (Jaccard 0.20).

eGFR and LV appear on both classic-selector and TabPFN sides. CaI is a TabPFN-train/SHAP name (means match Wang peak troponin I; still unnamed in the CSV). WBC is FDR-only because it was dropped from the ML matrix (Wang also excluded WBC from the Cox score). *k*-SII is one held-out VLST=1 row (cohort 5176). PDP is a train empirical prior near prevalence. None of this is a locked-in feature mask for nested CV, a causal stent-technique effect, or a monitoring rule.

The frozen Wang integer score recovers derivation-cohort ROC-AUC **0.8013** (published $c = 0.80$) with PR-AUC **0.1032**. Nested thinking v3.5 PR-AUC 0.9212 versus that integer score is a same-file comparison of foundation-model OOF against an 8-variable Cox point scale — **not** external validation and not inheritance of Wang’s Shantou test ($n = 2,058$, $c = 0.82$, file absent).

5.3 Methodological insight — subtle leakage in tabular cardiovascular ML

The first result that should change how this derivation file is modelled is not a TabPFN score. A published-style 70/30 GridSearch **overstates hold-out ranking** when follow-up time and related leak flags remain in the matrix (W1). ALL LEAKS ON versus OFF drops PR-AUC by **0.30–0.61** (LightGBM 0.9687 $\rightarrow$ 0.6675; logistic regression 0.9134 $\rightarrow$ 0.3431). Random forest F1 falls to **0**. Those arms are **not** “identical except TSSI”: five flags flip together, including SMOTE. Nested CV does not import those GridSearch winners and does not use SMOTE.

Incentives, as implemented. (1) TSSI as a covariate is binary-ified survival time (time-to-event vs completed follow-up). (2) WBC is a recording-precision / batch marker (and Wang excluded it from Cox). (3) Lab quantization is done **before** splitting so it is not a y -dependent encoder; a signature-only probe (243 indicators, AP 0.4270) shows how far decimal-grid membership alone can rank. Equalising leftover precision still drops TabPFN v3.5 nested AP by 0.0518. (4) Stent brands are encoded on the **train fold only** so rare strings cannot leak from the test distribution. (5) SMOTE is OFF except the leaks-on twin train set. Companion: identifiers dropped; scaler/OHE cloned inside splits; Part 2/5 catalogues do not mask nested CV. Follow-up drugs stay in with a post-baseline caveat. These are operational choices; they are not a claim that all possible leakage has been eliminated (brand-frequency leak on the ON arm is not separately quantified).

5.4 Limitations

1. **No external or temporal test of the ML models** (W3.1, B11). Nested CV on 5,185 derivation rows is not Shantou. Single-centre file (The First Hospital of Jilin University).
2. **Binary label versus Wang’s Cox time-to-event analysis** (W3.2). Dropping TSSI is required for honest binary classification and is not a Cox re-fit.

3. **Rare-event statistics / EPV** (W4): 92/81 $\approx$ **1.14**; Table 4 $\approx$ **5.4**; Table 4b $\approx$ **7.1** (still below EPV $\geq$ 10); 18–19 events per outer fold.
4. **Retrospective derivation-cohort design**; analysis not separately pre-registered.
5. Four TabPFN calibrations and two APIs must not be collapsed (W3.4–W3.5). Lowest nested ECE is local v3.5 **0.0003**; thinking v3.5 ECE **0.0028**. No decision-curve (B11) — nested PPV at an F1 cut is not clinical utility.
6. DAPT columns are post-baseline persistence (W3.6). WBC dual-label (W3.7). Unequal tuning (W3.8). Bootstrap does not re-fit (W3.9). LV / CaI unnamed (W3.10). Part 5 $\neq$ Part 4 predictor; SHAP dump records HTTP 429 then local finish (W3.11).
7. Interpretability was split because a full run exceeds the Kaggle session limit. SFS used **8** seeds, not the abandoned 10-seed plan. Client quota (20 thinking fits, 50M cells/day) constrained thinking-high design.
8. SMOTE only in ALL LEAKS ON. Single nested CV, not repeated. Cox LP / Shantou remain absent (B11).

These constraints bound the numbers: association under low EPV, attribution on a 70/30 split, and nested-CV discrimination on one derivation file.

5.5 Conclusion

On Wang 2020’s derivation cohort, after the full ALL LEAKS OFF protocol, TabPFN thinking v3.5 had the highest nested-CV PR-AUC among nine fitted arms (0.9212); local TabPFN v3.5 (0.8957) also exceeded library-default XGBoost (0.6322) and LightGBM (0.6271). That is not a hyperparameter-matched contest against tuned boosting. TSSI, WBC, raw lab decimals, a full-cohort stent codebook, and train SMOTE must not enter nested-CV binary classifiers as if they were baseline covariates. Recurring names in association and attribution (eGFR, LV, post-dilation flags, CaI on the TabPFN side) are not treatment effects. Nested-CV discrimination on this file is not external validation; the ML models were not externally or temporally tested.

Tables and figures

Markdown tables and figures only. Raw LaTeX is not used here (it does not render in the Markdown paper). Numbers from `paper/frozen_results.yaml`. Optimistic pooled F1 is **excluded**. Evidence-map §7.1 historical 0.8553 is **not** used.

Paths are relative to `manuscript/` so they display in `main_paper.md`.

Table 1. Cohort characteristics (association, not prediction)

Wang 2020 derivation file; $n = 5,185$ (no VLST 5,093; VLST 92). Continuous cells: mean (SD). Binary cells: n (%). Tests as in `eda.ipynb`. TSSI omitted (time-at-risk). DAPT is follow-up persistence. LV and CaI unnamed.

Variable	No VLST (n=5093)	VLST (n=92)	Test	p
Age, years	59.83 (9.93)	60.71 (11.33)	Welch t	0.463
LVEF, %	55.15 (4.52)	54.55 (3.68)	Mann-Whitney U	0.033
LV (unnamed; not in Wang Table 1)	44.55 (4.04)	49.11 (4.23)	Welch t	5.44e-17
WBC, 10 ⁹ /L	8.75 (3.24)	12.49 (3.92)	Mann-Whitney U	7.90e-21
Creatinine	72.53 (24.81)	72.44 (19.05)	Mann-Whitney U	0.879
eGFR	120.03 (34.10)	95.88 (19.63)	Welch t	4.64e-20
CaI (unnamed; not in Wang Table 1)	37.37 (61.64)	40.55 (72.25)	Mann-Whitney U	0.051
Fibrinogen, g/L	3.17 (0.88)	3.37 (1.01)	Mann-Whitney U	0.012
Stents per lesion	1.21 (0.46)	1.42 (0.65)	Mann-Whitney U	0.000
Total stent length, mm	31.70 (15.62)	36.46 (20.71)	Mann-Whitney U	0.001
Men	3489 (68.51%)	68 (73.91%)	Chi-square	0.268
Diabetes	1293 (25.39%)	36 (39.13%)	Chi-square	0.003
Hypertension	2670 (52.42%)	51 (55.43%)	Chi-square	0.567
Dyslipidaemia (HL)	1612 (31.65%)	28 (30.43%)	Chi-square	0.804
Current smoker	2860 (56.16%)	54 (58.70%)	Chi-square	0.626
Previous PCI	94 (1.85%)	10 (10.87%)	Fisher	1.25e-05
Previous MI	347 (6.81%)	10 (10.87%)	Chi-square	0.128
Admitting diagnosis AMI	3095 (60.77%)	65 (70.65%)	Chi-square	0.054
3-vessel disease	1422 (27.92%)	42 (45.65%)	Chi-square	0.000
SES (SES column; matches Wang Table 1 SES)	3502 (68.76%)	76 (82.61%)	Chi-square	0.004
1:1:1 post-dilation (CSV as stored)	2496 (49.01%)	14 (15.22%)	Chi-square	1.30e-10
No postdilation (CSV as stored)	2597 (50.99%)	78 (84.78%)	Chi-square	1.30e-10
eGFR < 90 (CKD90)	860 (16.89%)	32 (34.78%)	Chi-square	6.55e-06
DAPT during follow-up (not index PCI)	2260 (44.37%)	35 (38.04%)	Chi-square	0.226

Table 1. Cohort characteristics

Table 1. Restyle of paper_table_c_cohort_characteristics.png. Numeric copy:

Characteristic	No VLST (n = 5,093)	VLST (n = 92)	Test	p
Age, years	59.83 (9.93)	60.71 (11.33)	Welch	0.463
Men	3489 (68.51%)	68 (73.91%)	χ^2	0.268
Diabetes	1293 (25.39%)	36 (39.13%)	χ^2	0.003
Hypertension	2670 (52.42%)	51 (55.43%)	χ^2	0.567
Previous PCI	94 (1.85%)	10 (10.87%)	Fisher	1.25e-05
Previous MI	347 (6.81%)	10 (10.87%)	χ^2	0.128
Admitting diagnosis AMI	3095 (60.77%)	65 (70.65%)	χ^2	0.054
3-vessel disease	1422 (27.92%)	42 (45.65%)	χ^2	0.000
LVEF, %	55.15 (4.52)	54.55 (3.68)	MW	0.033
LV (unnamed)	44.55 (4.04)	49.11 (4.23)	Welch	5.44e-17
WBC, 10 ⁹ /L	8.75 (3.24)	12.49 (3.92)	MW	7.90e-21
Creatinine	72.53 (24.81)	72.44 (19.05)	MW	0.879
eGFR	120.03 (34.10)	95.88 (19.63)	Welch	4.64e-20
CaI (unnamed)	37.37 (61.64)	40.55 (72.25)	MW	0.051

Characteristic	No VLST (n = 5,093)	VLST (n = 92)	Test	p
Fibrinogen (Fiberinogen)	3.17 (0.88)	3.37 (1.01)	MW	0.012
Stents per lesion	1.21 (0.46)	1.42 (0.65)	MW	0.000
Total stent length, mm	31.70 (15.62)	38.46 (20.71)	MW	0.001
SES (PES column)	3502 (68.76%)	76 (82.61%)	χ^2	0.004
1.1:1 post-dilation (as stored)	2496 (49.01%)	14 (15.22%)	χ^2	1.30e-10
No postdilation (complement)	2597 (50.99%)	78 (84.78%)	χ^2	1.30e-10
eGFR < 90 (CKD90)	860 (16.89%)	32 (34.78%)	χ^2	6.55e-06
DAPT during follow-up	2260 (44.37%)	35 (38.04%)	χ^2	0.226

Table 2. Leakage contrast (ALL LEAKS ON vs ALL LEAKS OFF)

Stratified 70/30 hold-out (1,556 rows / 28 events). Seven classics, GridSearchCV. **Not nested CV. Not TabPFN.** Δ = ON – OFF. Gaussian NB ROC dual-labelled (higher OFF). RF F1 OFF = 0.

70/30 GridSearch: ALL LEAKS ON vs ALL LEAKS OFF

Model	With TSSI PR-AUC	Without TSSI PR-AUC	Δ PR-AUC	With TSSI ROC-AUC	Without TSSI ROC-AUC	With TSSI F1	Without TSSI F1
Logistic Regression	0.9134	0.3431	-0.5703	0.9954	0.8318	0.6753	0.2093
Decision Tree	0.7524	0.1378	-0.6146	0.9266	0.7009	0.7059	0.2791
Random Forest	0.9400	0.4874	-0.4526	0.9977	0.9287	0.8333	0.0000
Gaussian NB	0.2728	0.0564	-0.2163	0.7425	0.7493	0.0437	0.0370
CatBoost	0.9599	0.4942	-0.4656	0.9964	0.9103	0.9231	0.4096
XGBoost	0.9547	0.5685	-0.3862	0.9963	0.9183	0.9231	0.5500
LightGBM	0.9687	0.6675	-0.3011	0.9986	0.9404	0.9231	0.4444

Table 2. Leakage metrics

Figure S-TSSI. PR-AUC collapse on the 1,556-row hold-out. Dotted line = prevalence 0.0177.

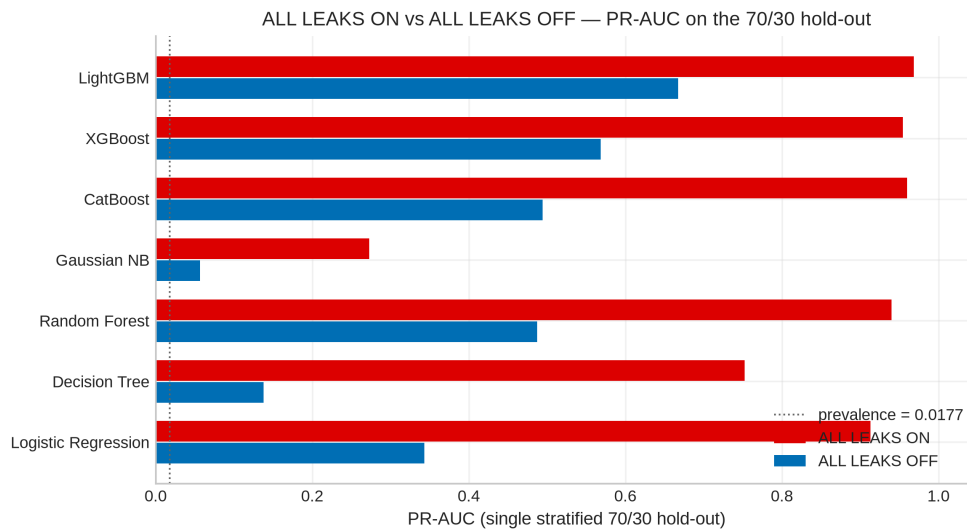


Figure S-TSSI. PR-AUC ON vs OFF

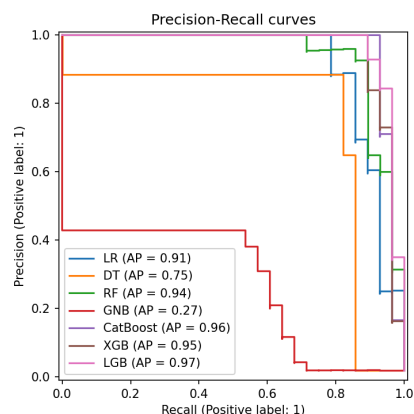
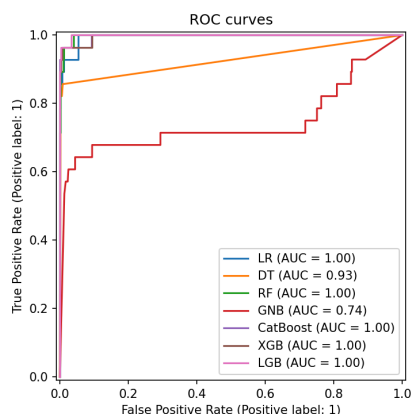
2a. PR-AUC

Model	ON	OFF	Δ
Logistic regression	0.9134	0.3431	+0.5703
Decision tree	0.7524	0.1378	+0.6146
Random forest	0.9400	0.4874	+0.4526
Gaussian NB	0.2728	0.0564	+0.2163
CatBoost	0.9599	0.4942	+0.4656
XGBoost	0.9547	0.5685	+0.3862
LightGBM	0.9687	0.6675	+0.3011

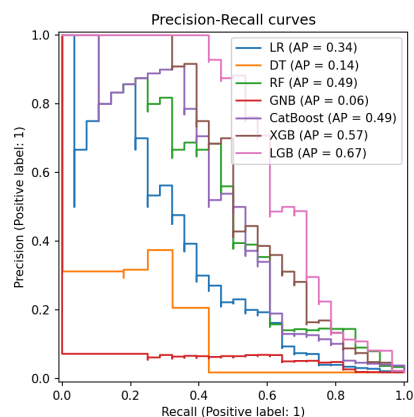
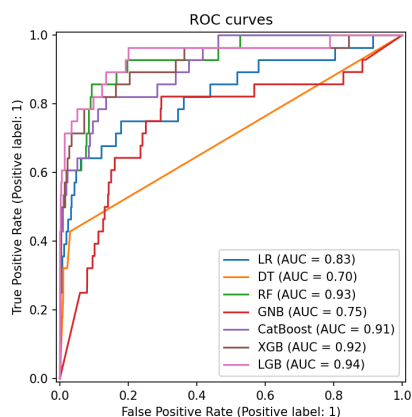
2b. F1 at the notebook cut

Model	F1 ON	F1 OFF
Logistic regression	0.6753	0.2093
Decision tree	0.7059	0.2791
Random forest	0.8333	0.0000
Gaussian NB	0.0437	0.0370
CatBoost	0.9231	0.4096
XGBoost	0.9231	0.5500
LightGBM	0.9231	0.4444

Dump ROC/PR curves:



ALL LEAKS ON ROC/PR



ALL LEAKS OFF ROC/PR

Table 3. Nested-CV benchmark (anti-leakage ON, 9 arms)

Pooled OOF ranking + honest nested operating point (inner-fold F1 thresholds). Anti-leakage ON = ALL LEAKS OFF. Five classic rows are **library defaults + class weights**, not nested GridSearch. Bootstrap 95% CIs on PR-AUC ($n_{boot}=2000$, seed 42). Pins `tabpfn==9.0.0 / tabpfn-client==0.6.0`. Do **not** quote pooled F1 cuts instead of nested Table 2.

Figure 1. Nested-CV out-of-fold PR (left) and ROC (right).

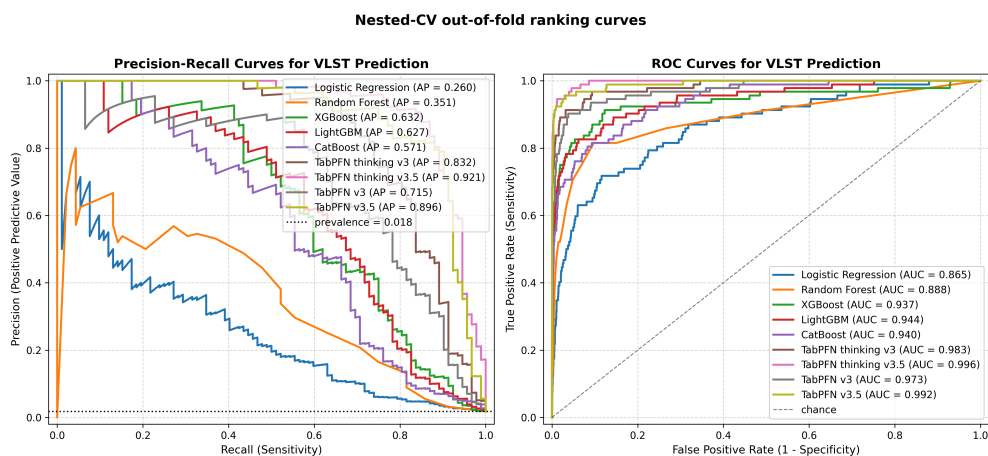


Figure 1. Nested OOF PR and ROC

Figure 2. Nested-CV calibration (quantile bins).

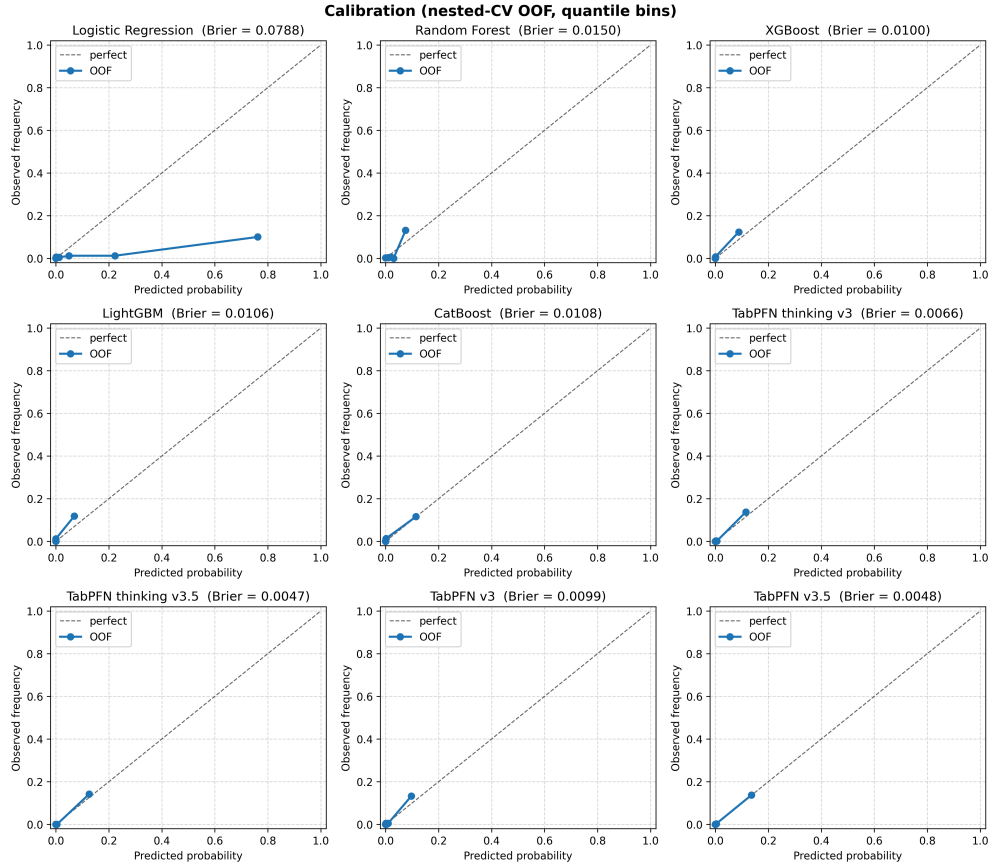


Figure 2. Nested calibration

Pooled nested-CV OOF ranking after anti-leakage (n=5185, 92 events)

Rank	Model	PR-AUC	PR fold mean $\pm$ SD	ROC-AUC	ROC fold mean $\pm$ SD	Brier
1	TabPFN thinking v3.5	0.9212	0.9227 $\pm$ 0.0423	0.9963	0.9955 $\pm$ 0.0027	0.0047
2	TabPFN v3.5	0.8957	0.8995 $\pm$ 0.0528	0.9916	0.9922 $\pm$ 0.0066	0.0048
3	TabPFN thinking v3	0.8319	0.8335 $\pm$ 0.0658	0.9834	0.9843 $\pm$ 0.0105	0.0066
4	TabPFN v3	0.7350	0.7286 $\pm$ 0.0319	0.9731	0.9741 $\pm$ 0.0125	0.0099
5	XGBoost	0.6322	0.6448 $\pm$ 0.0880	0.9374	0.9366 $\pm$ 0.0371	0.0100
6	LightGBM	0.6273	0.6248 $\pm$ 0.0990	0.9444	0.9458 $\pm$ 0.0332	0.0106
7	CatBoost	0.5707	0.5867 $\pm$ 0.0494	0.9404	0.9418 $\pm$ 0.0188	0.0108
8	Random Forest	0.3506	0.3843 $\pm$ 0.0792	0.8883	0.8880 $\pm$ 0.0228	0.0150
9	Logistic Regression	0.2596	0.2836 $\pm$ 0.1106	0.8651	0.8671 $\pm$ 0.0479	0.0788

Ranking metrics table

Honest nested-CV F1 operating point (inner-fold threshold on unseen outer fold)

Model	Threshold (mean $\pm$ SD)	Accuracy	Precision	Recall	Specificity	NPV	F1	F2	TN	FP	FN	TP
TabPFN thinking v3.5	0.318 $\pm$ 0.059	0.9952	0.8951	0.8370	0.9980	0.9973	0.8903	0.8402	5083	10	15	77
TabPFN v3.5	0.386 $\pm$ 0.061	0.9940	0.8675	0.7626	0.9978	0.9961	0.8229	0.7982	5082	11	20	72
TabPFN thinking v3	0.271 $\pm$ 0.055	0.9829	0.8767	0.6957	0.9982	0.9945	0.7758	0.7256	5084	9	28	64
TabPFN v3	0.183 $\pm$ 0.036	0.9886	0.7089	0.6087	0.9955	0.9920	0.6550	0.6264	5070	23	36	56
XGBoost	0.238 $\pm$ 0.104	0.9867	0.6575	0.5217	0.9951	0.9914	0.5818	0.5442	5068	25	44	48
LightGBM	0.076 $\pm$ 0.037	0.9855	0.5834	0.5870	0.9927	0.9925	0.5902	0.5882	5056	37	38	54
CatBoost	0.170 $\pm$ 0.064	0.9828	0.5093	0.5978	0.9886	0.9927	0.5500	0.5777	5040	53	37	55
Random Forest	0.112 $\pm$ 0.016	0.9801	0.4421	0.4565	0.9896	0.9902	0.4492	0.4536	5040	53	50	42
Logistic Regression	0.964 $\pm$ 0.013	0.9740	0.3028	0.3587	0.9851	0.9884	0.3284	0.3459	5017	76	59	33

Honest nested operating point

Model	PR-AUC (95% CI)	ROC-AUC (95% CI)	Brier (95% CI)
Logistic Regression	0.2596 [0.1834, 0.3588]	0.8651 [0.8220, 0.9044]	0.0788 [0.0730, 0.0844]
Random Forest	0.3506 [0.2660, 0.4633]	0.8883 [0.8413, 0.9296]	0.0150 [0.0144, 0.0155]
XGBoost	0.6322 [0.5331, 0.7247]	0.9374 [0.8967, 0.9699]	0.0100 [0.0084, 0.0116]
LightGBM	0.6271 [0.5313, 0.7202]	0.9444 [0.9135, 0.9698]	0.0106 [0.0090, 0.0123]
CatBoost	0.5707 [0.4750, 0.6729]	0.9404 [0.9142, 0.9632]	0.0108 [0.0091, 0.0125]
TabPFN thinking v3	0.8319 [0.7626, 0.8945]	0.9834 [0.9709, 0.9939]	0.0066 [0.0053, 0.0080]
TabPFN thinking v3.5	0.9212 [0.8785, 0.9613]	0.9963 [0.9932, 0.9986]	0.0047 [0.0037, 0.0058]
TabPFN v3	0.7150 [0.6260, 0.8074]	0.9731 [0.9551, 0.9883]	0.0099 [0.0089, 0.0111]
TabPFN v3.5	0.8957 [0.8446, 0.9426]	0.9916 [0.9828, 0.9978]	0.0048 [0.0038, 0.0060]

Bootstrap 95% CIs

Model	PR-AUC [95% CI]	ROC-AUC	Brier	Sensitivity	Precision	F1
TabPFN thinking v3.5	0.9212 [0.8785, 0.9613]	0.9963	0.0047	0.8370	0.8851	0.8603
TabPFN v3.5	0.8957 [0.8446, 0.9426]	0.9916	0.0048	0.7826	0.8675	0.8229
TabPFN thinking v3	0.8319 [0.7626, 0.8945]	0.9834	0.0066	0.6957	0.8767	0.7758
TabPFN v3	0.7150 [0.6260, 0.8074]	0.9731	0.0099	0.6087	0.7089	0.6550
XGBoost	0.6322 [0.5331, 0.7247]	0.9374	0.0100	0.5217	0.6575	0.5818
LightGBM	0.6271 [0.5313, 0.7202]	0.9444	0.0106	0.5870	0.5934	0.5902
CatBoost	0.5707 [0.4750, 0.6729]	0.9404	0.0108	0.5978	0.5093	0.5500
Random forest	0.3506 [0.2660, 0.4633]	0.8883	0.0150	0.4565	0.4421	0.4492
Logistic regression	0.2596 [0.1834, 0.3588]	0.8651	0.0788	0.3587	0.3028	0.3284

Thinking v3.5 nested 2×2: 5083/10/15/77. TabPFN v3.5: 5082/11/20/72. LightGBM: 5056/37/38/54. Frozen Wang integer score (not a nested arm): ROC-AUC 0.8013, PR-AUC 0.1032.

Table 4. Eight-seed forward SFS rankings (train, local TabPFN v3.5)

Keep 10 of 80 columns; 5-fold CV; average precision; STABILITY_N_SEEDS=8. Train n = 3,629. WBC dropped. Not a nested-CV feature mask.

Stability selection frequency (8/8 seeds × top-10 forward SFS; train)

Feature	Selected	Frequency
CaI	8/8	1.000
LV	8/8	1.000
eGFR	8/8	1.000
Cre	7/8	0.875
HbA1c	7/8	0.875
Age	6/8	0.750
LVEF	5/8	0.625
1.1:1Post dilation	4/8	0.500
LDL	2/8	0.250
No postdilation	2/8	0.250
TCL	2/8	0.250
Fibrinogen	1/8	0.125
Initial diagnosis-AMI	1/8	0.125
Previous PCI	1/8	0.125
Slow flow	1/8	0.125
Stent type-SES	1/8	0.125

Table 4. SFS stability

Rank	Feature	Selected	Frequency
1	CaI	8/8	1.000
2	LV	8/8	1.000
3	eGFR	8/8	1.000
4	Cre	7/8	0.875
5	HbA1c	7/8	0.875
6	Age	6/8	0.750
7	LVEF	5/8	0.625
8	1.1:1Post dilation	4/8	0.500
9	LDL	2/8	0.250
10	No postdilation	2/8	0.250
11	TCL	2/8	0.250
12	Fibrinogen	1/8	0.125
13	Initial diagnosis-AMI	1/8	0.125
14	Previous PCI	1/8	0.125
15	Slow flow	1/8	0.125

Rank	Feature	Selected	Frequency
16	Stent type-SES	1/8	0.125

Held-out mean(|SHAP|) for the 8/8 names (1,556 rows): eGFR 1.2288, CaI 1.0867, LV 0.4828 (Cre 0.8093 is 7/8 SFS).

Interpretability figures

Mutual information (train).

Top 15 mutual information (train n=3629; 80-column dump; WBC dropped)

Rank	Feature	Mutual information
1	CaI	0.020536
2	LV	0.012818
3	eGFR	0.009424
4	LDL	0.009367
5	HbA1c	0.007328
6	Lesion location-Ostial	0.005293
7	P-RCA	0.004830
8	HGB	0.004365
9	1.1:1Post dilation	0.004057
10	HDL	0.003823
11	Fibrinogen	0.003712
12	Current drinking	0.003202
13	Stent type-SES	0.003176
14	stent overlap	0.003081
15	P-LCX	0.003014

Table. Top 15 mutual information

Figure. Continuous PDP (train; empirical prior; not nested-CV risk).

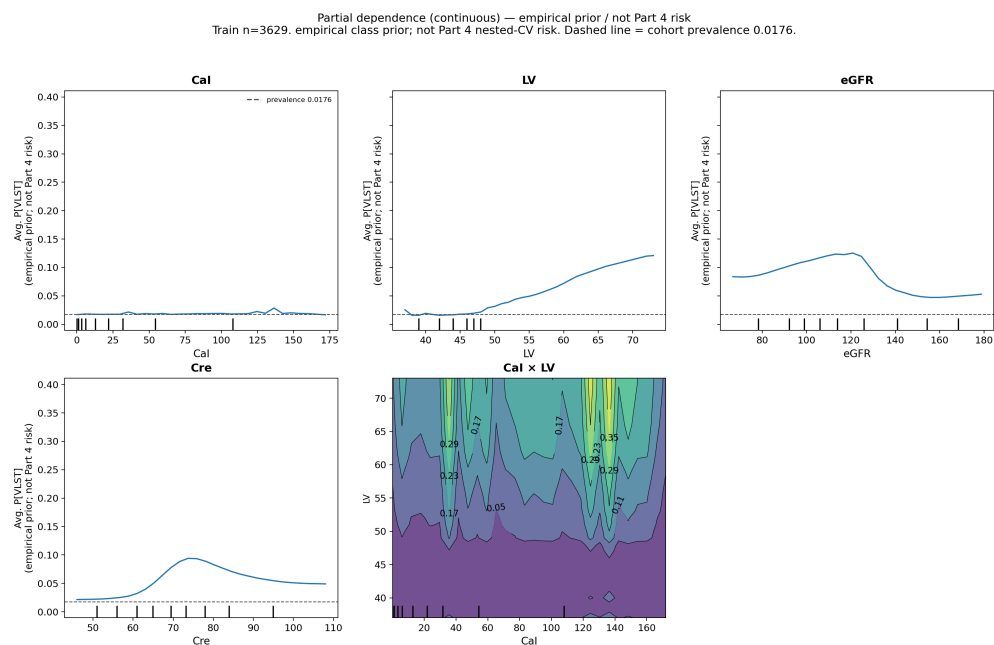


Figure. Continuous PDP

Figure. Binary PDP.

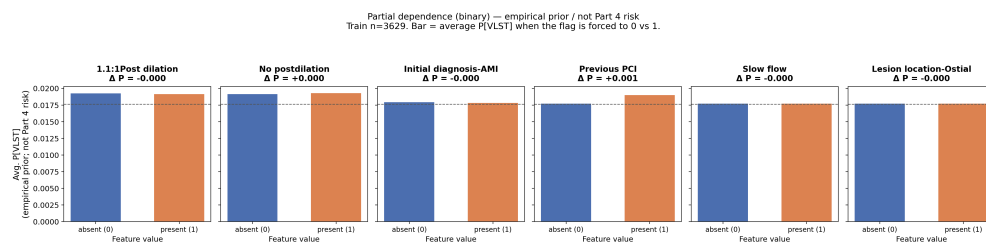


Figure. Binary PDP

Figure. SHAP summary (1,556 held-out rows).

SHAP summary — feature impact on P[Stent thrombosis] (held-out, n=1556)

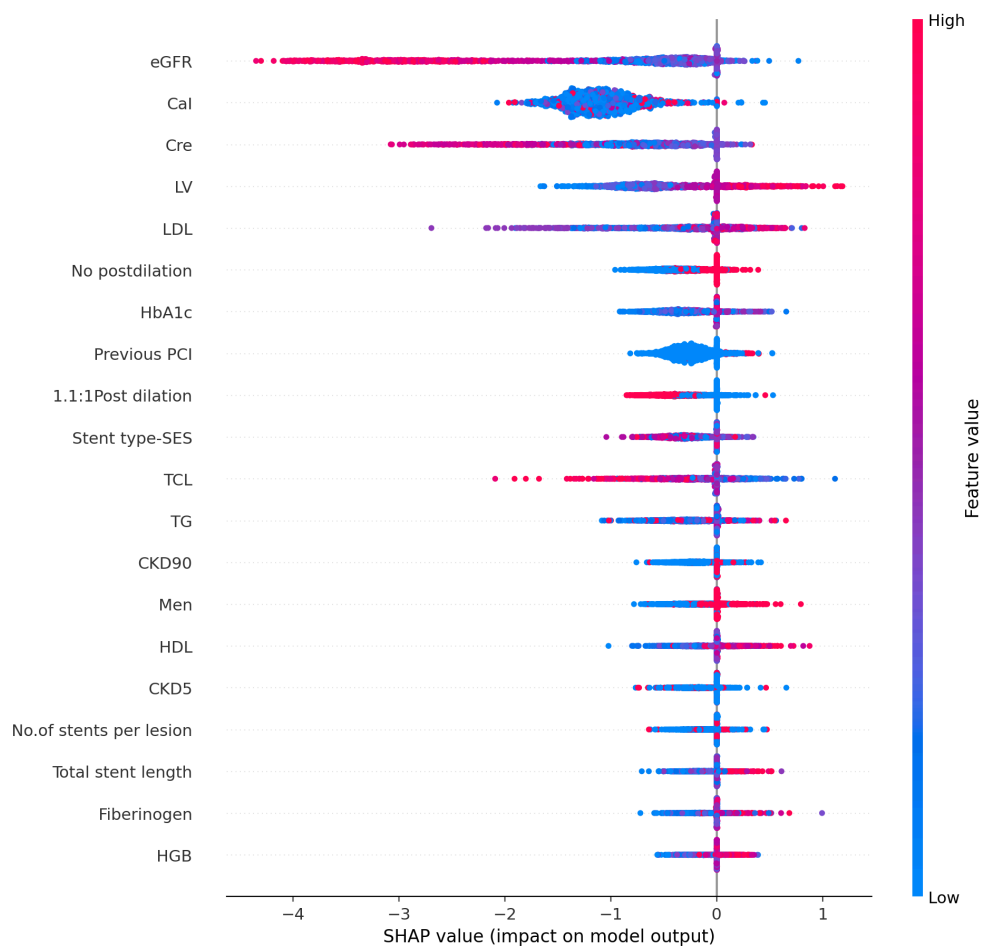


Figure. SHAP summary

Figure. Mean |SHAP| bar. Leading: eGFR 1.2288, Cal 1.0867, Cre 0.8093, LV 0.4828.

SHAP bar — mean(|SHAP|) on the held-out split (n=1556)

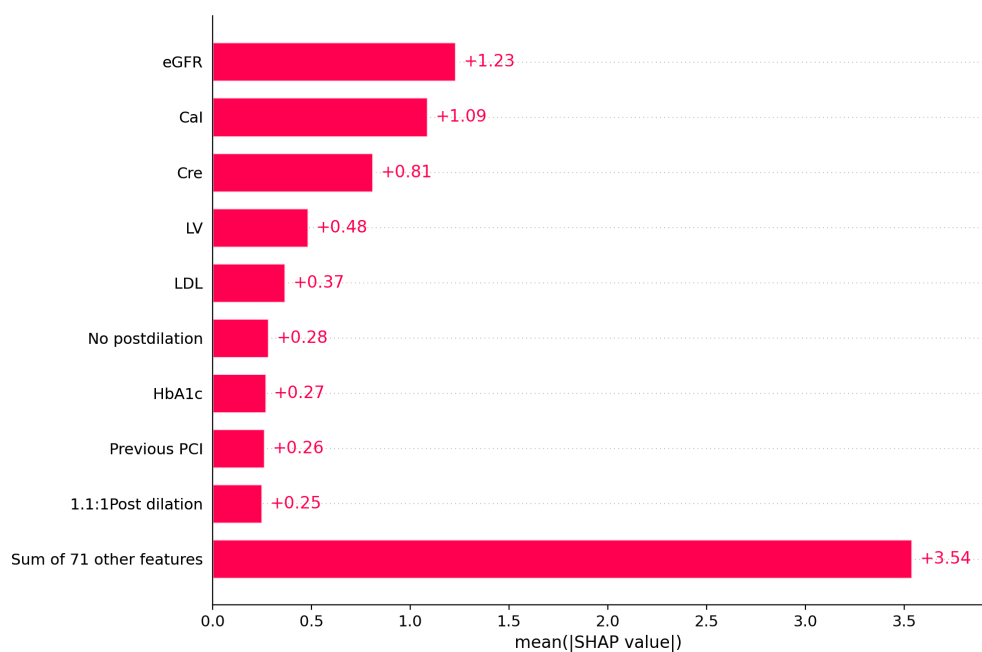


Figure. SHAP bar

Figure. SHAP beeswarm.

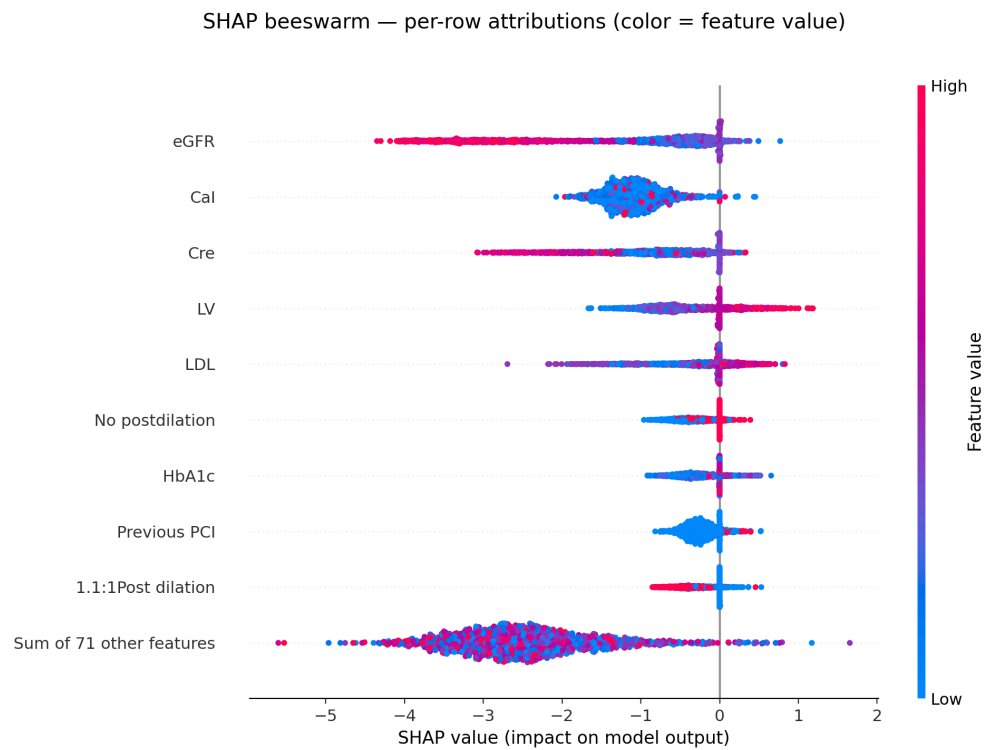


Figure. SHAP beeswarm

Figure. One-row waterfall (cohort row 5176).

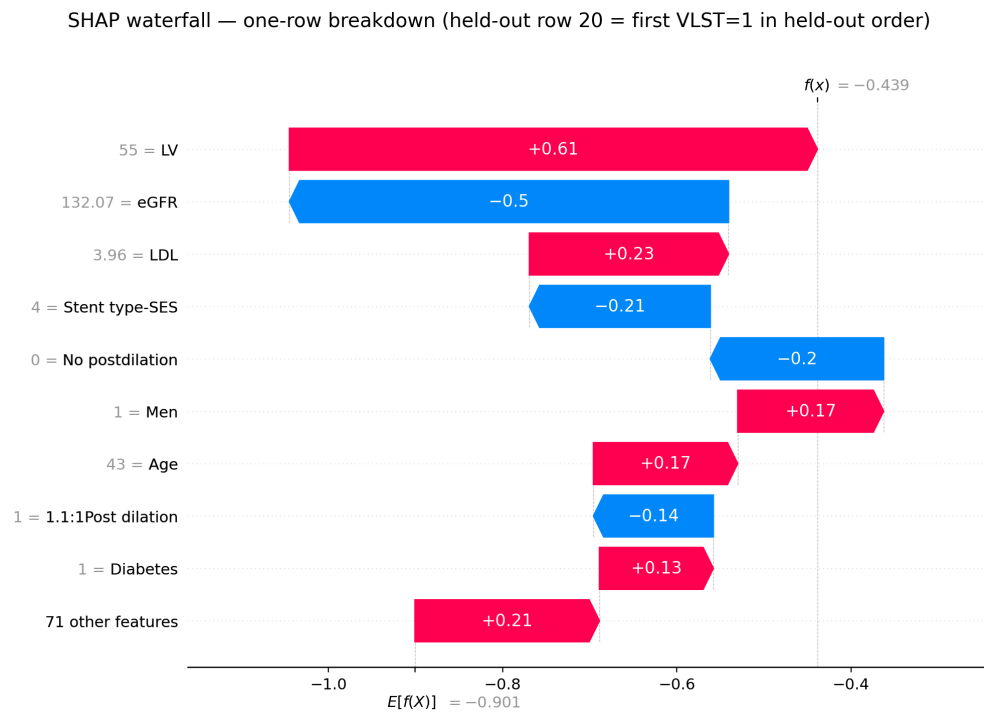


Figure. SHAP waterfall

Figure. *k*-SII network (same one VLST=1 row; not a cohort interaction screen).

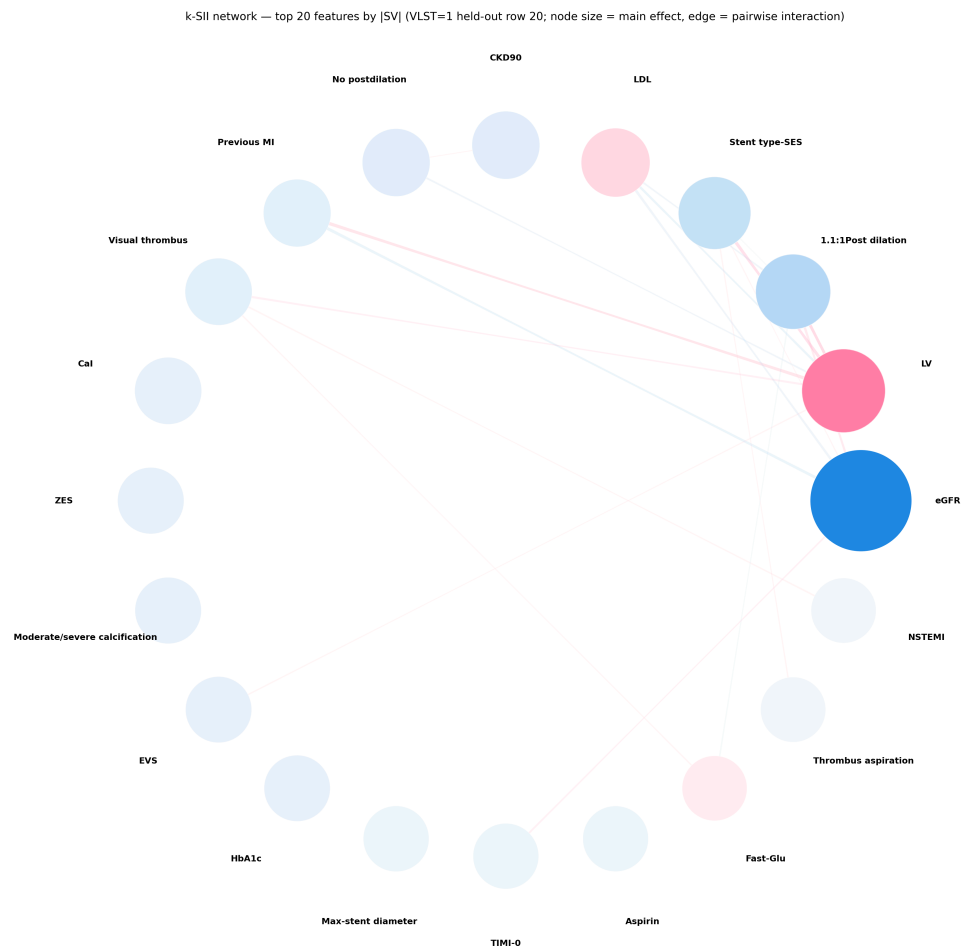


Figure. Consensus ranking (train MI + train SFS + held-out SHAP).

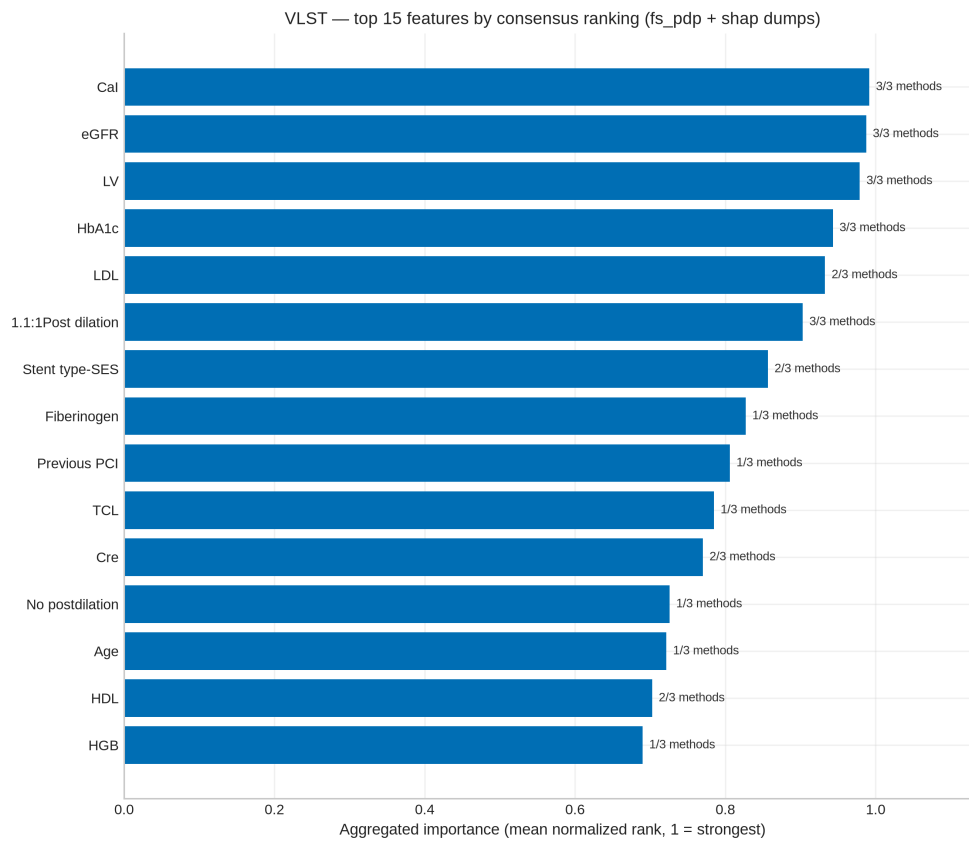


Figure. Consensus ranking

Borda consensus (train MI + train SFS + held-out mean|SHAP; 80 columns)

Rank	Feature	Score	n methods	Stability	mean(SHAP)	MI	In MI top	In SHAP top
1	Cal	0.9916	3/3	1.000	1.0867	0.020536	yes	yes
2	eGFR	0.9873	3/3	1.000	1.2288	0.009424	yes	yes
3	LV	0.9789	3/3	1.000	0.4828	0.012818	yes	yes
4	HbA1c	0.9430	3/3	0.875	0.2673	0.007328	yes	yes
5	LDL	0.9325	2/3	0.250	0.3659	0.009367	yes	yes
6	1.1:1Post dilation	0.9030	3/3	0.500	0.2483	0.004057	yes	yes
7	Stent type-SES	0.8565	2/3	0.125	0.2402	0.003176	yes	yes
8	Fibrinogen	0.8270	1/3	0.125	0.0694	0.003712	yes	no
9	Previous PCI	0.8059	1/3	0.125	0.2604	0.001986	no	yes
10	TCL	0.7848	1/3	0.250	0.2369	0.001167	no	yes
11	Cre	0.7700	2/3	0.875	0.8093	0.000338	no	yes
12	No postdilation	0.7257	1/3	0.250	0.2812	0.000079	no	yes
13	Age	0.7215	1/3	0.750	0.0373	0.002065	no	no
14	HDL	0.7025	2/3	0.000	0.1174	0.003823	yes	yes
15	HGB	0.6899	1/3	0.000	0.0644	0.004365	yes	no

Table. Consensus report

Do not quote pooled confusion matrices (paper_fig3_confusion_matrices.png) as nested operating points.